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Expressive power and succinctness of the positive calculus of binary relations ☆

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ABSTRACT

In this paper, we study the expressive power and succinctness of *the positive calculus of (binary) relations*. We show that, for binary relations, (1) the calculus has the same expressive power as that of three-variable existential positive (first-order) logic, and (2) the calculus is exponentially less succinct than three-variable existential positive logic, namely, there is no subexponential-size translation from three-variable existential positive logic to the calculus. Additionally, we give a more fine-grained expressive power equivalence between the (full) calculus of relations and three-variable first-order logic in terms of the quantifier alternation hierarchy. It remains open whether the calculus of relations is also exponentially less succinct than three-variable first-order logic.

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1. Introduction

The calculus of (binary) relations (denoted by CoR, for short), which was revived by Tarski [2], is an algebraic system on binary relations. The calculus of relations and relation algebras have many applications in various areas of computer science, e.g., databases, program development and verification, and program semantics (see [3] for more details and references). We can simply express some properties of binary relations by using (in)equational formulas of CoR; for example, the formula $a \cdot a \leq a$ indicates that the binary relation a is transitive. Here, the symbol $\cdot$ denotes the composition operator of binary relations, and the symbol $\leq$ denotes the inclusion relation on sets. CoR has a certain high expressive power, namely the expressive power of CoR is equivalent to *three-variable first-order logic* (denoted by FO^3) in terms of binary relations [4–6]. One of the downsides for this high expressive power is that the equational theory of CoR is undecidable [4], even for terms built only from one variable, union, complement, and composition [7].

In this paper, we focus on *the positive calculus of relations* (denoted by PCoR, for short) [8,9], which is a complement-free fragment of CoR. Namely, PCoR terms are built from union, intersection, composition, converse, the identity relation, the empty relation, and the universal relation. PCoR is strictly less expressive than CoR, but its equational theory is decidable [8]. This decidability result also holds when adding a transitive closure operation, thus arriving at Kleene allegory terms [10].

The first contribution of this paper is to show that PCoR has the same expressive power as *three-variable existential positive (first-order) logic* (denoted by EP^3) in terms of binary relations (Theorem 3.1). The standard translation from CoR to FO^3 [2, pp. 75-76] naturally specializes into a translation from PCoR to EP^3. Conversely, translations from FO^3 to CoR also

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exist [4, Sect. 3.9][5, Sect. 20][6, Thm. 552], but they generate non-PCoR terms on the EP^3 fragment. Hence, we have to refine them. The translation we propose uses disjunctive/conjunctive normal forms where literals have at most two free variables. We define it from full FO^3 into CoR by relying on a relational sum operation (†), which is dual to composition ($\cdot$). By specializing our translation to the various considered fragments, we obtain (1) an exponential-size translation from EP^3 to PCoR, and (2) a perfect match between the quantifier alternation hierarchy in FO^3 and *dot-dagger* ($\cdot$-†) *alternation hierarchy* in CoR. Roughly speaking, it shows that the two operators, $\cdot$ and †, from the calculus of relations, exactly correspond to the two quantifiers, ∃ and ∀, from first-order logic.

The second contribution of this paper is to show that PCoR is exponentially less succinct than EP^3 (Theorem 5.1). Namely, the exponential blowup in translating from EP^3 to PCoR is unavoidable. Hence, in this sense, the exponential-size translation for EP^3 given in this paper is tight. Our proof in this paper is proceeded by using weight functions based on *formula-size games* [11–14] (or called Adler-Immerman games), which is a variant of Ehrenfeucht-Fraïssé games [15,16] (see also [17, Sect. 2]). We introduce a formula-size game for CoR (Section 4).

Furthermore, we extend the two above results for both transitive closure extensions, namely, we show that PCoR with transitive closure [9] (denoted by PCoR(TC)) has the same expressive power as EP^3 with *variable-confined monadic transitive closure* (denoted by EP^3(v-MTC)) and that PCoR(TC) is exponentially less succinct than EP^3(v-MTC).

Remark 1.1 *(On the trade-off between succinctness and tractability). The (binary relation) query evaluation problem* [18] (of the combined complexity) is the problem to decide for a structure M, a term t, and a pair of nodes in M, whether the pair is in the binary relation denoted by t on the structure M. While PCoR is strictly less succinct than EP^3, PCoR is more tractable than EP^3 in the simple dynamic-programming algorithm for this problem (see, e.g., [19, Prop. 6.6]). While it does not imply a certain computational complexity gap between the two problems, we obtain an $\mathcal{O}(\|t\| \times \|M\|^2)$-time algorithm for PCoR if the number of occurrences of $\cdot$ is fixed, whereas we only obtain an $\mathcal{O}(\|t\| \times \|M\|^3)$-time algorithm for EP^3 even if the number of occurrences of ∃ is fixed. Here, $\|t\|$ is the size of t and $\|M\|$ is the cardinality of the domain of M, respectively.

Related work

Expressive power of formal systems is widely studied in mathematical logic and computer science. An example is that the following systems have the same expressive power in terms of *recognizability over word structures*: regular expressions, deterministic finite automata, non-deterministic finite automata, and monadic second-order logic (MSO) (see, e.g., [17, Sect. 6]). However, these four systems are certainly different in terms of *succinctness*. For example, while there is an (exponential-size) translation from non-deterministic finite automata to deterministic finite automata by the powerset construction, there is no polynomial-size translation [20, Prop. 1]. This can yield significant complexity differences for various problems, e.g., membership, universality, and equivalence testing have different complexity depending on whether we start from a term, an automaton, or a formula. See [21, Thm. 16] and [22, Thm. 11] for the other succinctness gaps among regular expressions, deterministic finite automata, and non-deterministic finite automata. The succinctness gap between MSO and each of the other three systems can be shown by the following fact [23] obtained from Stockmeyer's proof [24]: the minimum length of strings that satisfy a given FO formula is non-elementary in the size of formula. (On the other hand, the length is at most exponential for the other three systems.) Additionally, the expressive power equivalence like the above is known for FO. The following have the same expressive power: star-free regular expressions, FO^3, and FO (see, e.g., [17, Sect. 6]). In [12], it is shown that FO^3 is exponentially less succinct than FO (precisely, FO^4) over unary alphabet words. Another example is that the following classes of formulas have the same expressive power with respect to *boolean queries*: propositional logic formulas, negation normal form formulas, and disjunctive/conjunctive normal form formulas. In [25], the succinctness among a dozen formula classes (including the above ones) is investigated. Furthermore, between modal logic and FO, it is shown in [14] that (basic) modal logic (and modal μ-calculus) is non-elementary less succinct than bisimulation invariant FO.

In this paper, we compare the succinctness between PCoR and EP^3. To the best of our knowledge, it is the first comparison between the succinctness of the (positive) calculus of relations and those of other systems.

Organization

Section 2 provides the definitions of CoR and FO^3 and fragments of them (including PCoR and EP^3), the notions of the expressive power and succinctness, and the standard translation from CoR to FO^3. Section 3 presents a new translation from FO^3 to CoR. Consequently, we have that PCoR has the same expressive power as EP^3. Section 4 introduces a *formula-size game* for CoR. Section 5 shows that PCoR is exponentially less succinct than EP^3. Section 6 presents some extensions of the succinctness result in Section 5. Section 7 extends the results in Section 3-5 by adding a transitive closure operator. Section 8 concludes this paper.

In comparison with the conference version [1], the first main theorem (Theorem 3.1) is improved; the case of FO^2 has been added in Section 3.2; we introduce a formula-size game for CoR in Section 4; we present a more robust proof for succinctness in Section 5 (Theorem 5.1) using the formula-size game; we present some extensions of Theorem 5.1 in Section 6.

2. Preliminaries

$\mathbb{N}$ (resp. $\mathbb{N}_+$) denotes the set of all non-negative (resp. positive) integers. For $l, r \in \mathbb{N}$ such that $l \leq r$, $[l, r]$ denotes the set $\{m \in \mathbb{N} \mid l \leq m \leq r\}$ and $[n]$ denotes the set $[1, n]$. $\#(A)$ denotes the cardinality of a set A.

Let $\mathscr{A}$ be a set of *binary relation symbols*. A *structure M (of binary relations)* is a tuple $\langle |M|, \{a^M\}_{a\in\mathscr{A}}\rangle$, where $|M|$ is a non-empty set, and for each $a \in \mathscr{A}$, $a^M \subseteq |M|^2$ is a binary relation on $|M|$. A *k-pointed structure $\tilde{M}$ (of binary relations)* is a tuple $\langle M^{\tilde{M}}, v_1^{\tilde{M}}, \ldots, v_k^{\tilde{M}}\rangle$, where $M^{\tilde{M}}$ is a structure (we may denote $M^{\tilde{M}}$ by $\tilde{M}$, for notational simplicity) and $v_1^{\tilde{M}}, \ldots, v_k^{\tilde{M}} \in |\tilde{M}|$. We will use the notions for 0-pointed structures $\langle M\rangle$ also for structures M. We denote by $\mathbf{M}$ (resp. $\mathbf{M}_k$) the set of all structures (resp. all k-pointed structures). Let $\tilde{M}$ and $\tilde{N}$ be k-pointed structures and let h be a function from $|\tilde{M}|$ to $|\tilde{N}|$. Then, we say that h is a *homomorphism*, written $h\colon \tilde{M} \to \tilde{N}$ if for every $a \in \mathscr{A}$, $v, w \in |\tilde{M}|$, $\langle v, w\rangle \in a^{\tilde{M}} \Longrightarrow \langle h(v), h(w)\rangle \in a^{\tilde{N}}$, and for every $i \in [k]$, $h(v_i^{\tilde{M}}) = v_i^{\tilde{N}}$; h is a *strong homomorphism* if h is a homomorphism, and for every $a \in \mathscr{A}$, $v, w \in |\tilde{M}|$, $\langle h(v), h(w)\rangle \in a^{\tilde{N}} \Longrightarrow \langle v, w\rangle \in a^{\tilde{M}}$; h is an *embedding*, written $h\colon \tilde{M} \subseteq \tilde{N}$ if h is an injective strong homomorphism; h is an *isomorphism*, written $h\colon \tilde{M} \cong \tilde{N}$ if h is a bijective strong homomorphism. We say that $\tilde{M}$ is *homomorphic* to $\tilde{N}$ (resp. a *submodel* of $\tilde{N}$, *isomorphic* to $\tilde{N}$), written $\tilde{M} \to \tilde{N}$ (resp. $\tilde{M} \subseteq \tilde{N}$, $\tilde{M} \cong \tilde{N}$) if there exists a homomorphism (resp. embedding, isomorphism) from $\tilde{M}$ to $\tilde{N}$.

For a set A and an equivalence relation $\sim$ on A, we denote by $A/\!\sim$ the quotient set of A by $\sim$ and denote by $[a]_\sim$ the equivalence class of an element a in A. For a nonempty sequence $\langle A_i\rangle_{i\in[k]}$ of sets, we denote by $\biguplus_{i\in[k]} A_i$ the set $\{\langle i, a_i\rangle \mid i \in [k], a_i \in A_i\}$. We may write $A_1 \uplus A_2$ for $\biguplus_{i\in[2]} A_i$.

For a structure M and an equivalence relation $\sim$ on $|M|$, we denote by $M/\!\sim$ the structure defined by $|M/\!\sim| := |M|/\!\sim$ and $a^{M/\sim} := \{\langle [v]_\sim, [w]_\sim\rangle \mid$ there is some $\langle v', w'\rangle \in [v]_\sim \times [w]_\sim$ such that $\langle v', w'\rangle \in a^M\}$. For a nonempty sequence $\langle M_i\rangle_{i\in[k]}$ of structures, we denote by $\biguplus_{i\in[k]} M_i$ the structure defined by $|\biguplus_{i\in[k]} M_i| := \biguplus_{i\in[k]} |M_i|$ and $a^{\biguplus_{i\in[k]} M_i} := \{\langle\langle i, v\rangle, \langle i, w\rangle\rangle \mid i \in [k], \langle v, w\rangle \in a^{M_i}\}$.

The calculus of relations and its fragments

We introduce the calculus of relations (CoR) [2] and its syntactic fragments: *the positive calculus of relations* (PCoR) [8,9] and *the primitive positive calculus of relations* (denoted by PPCoR, a.k.a. allegory terms with top [26]). The terms of CoR consist of the following basic operations on binary relations. Let X be a set. For two binary relations R and S on the universe X, the *union* $R \cup S$, *intersection* $R \cap S$, and *complement* R^- are defined as the corresponding set-theoretic operators, respectively. The symbols $\mathbf{0}$ and $\top$ are employed to denote the *empty relation* and the *universal relation*, respectively. *Relational composition* (a.k.a. *relational multiplication*) $R \cdot S$ is defined as $\{\langle v, v'\rangle \in X^2 \mid \exists w.\langle v, w\rangle \in R \wedge \langle w, v'\rangle \in S\}$, and *relational sum* $R \dagger S$ is defined as $\{\langle v, v'\rangle \in X^2 \mid \forall w.\langle v, w\rangle \in R \vee \langle w, v'\rangle \in S\}$. In this paper, the *projection* R^π is defined as $\{\langle v_1, v_2\rangle \in X^2 \mid \langle v_{\pi(1)}, v_{\pi(2)}\rangle \in R\}$ for each function $\pi\colon [2] \to [2]$. The symbol $\mathbf{1}$ is employed to denote the *identity relation*. We now define the syntax and semantics of CoR. The set of *terms* of CoR/PCoR/PPCoR is given by the following grammar, where $a \in \mathscr{A}$:

$$t, s \in \mathrm{Term}^{\mathrm{CoR}} ::= t^\pi \mid a \mid \mathbf{1} \mid \top \mid t \cap s \mid t \cdot s \mid \mathbf{0} \mid t \cup s \mid t \dagger s \mid t^-$$
$$t, s \in \mathrm{Term}^{\mathrm{PCoR}} ::= t^\pi \mid a \mid \mathbf{1} \mid \top \mid t \cap s \mid t \cdot s \mid \mathbf{0} \mid t \cup s$$
$$t, s \in \mathrm{Term}^{\mathrm{PPCoR}} ::= t^\pi \mid a \mid \mathbf{1} \mid \top \mid t \cap s \mid t \cdot s$$

We use $t^\smile$ (the *converse* of t) to denote $t^{\{1\mapsto 2, 2\mapsto 1\}}$. The *semantics* $[\![t]\!]_M \subseteq |M|^2$ of a CoR term t on a structure M is a binary relation on $|M|$, which is defined by: $[\![t^\pi]\!]_M := [\![t]\!]_M^\pi$; $[\![a]\!]_M := a^M$; $[\![\mathbf{1}]\!]_M := \triangle(|M|)$; $[\![\top]\!]_M := |M|^2$; $[\![t \cap s]\!]_M := [\![t]\!]_M \cap [\![s]\!]_M$; $[\![t \cdot s]\!]_M := [\![t]\!]_M \cdot [\![s]\!]_M$; $[\![\mathbf{0}]\!]_M := \emptyset$; $[\![t \cup s]\!]_M := [\![t]\!]_M \cup [\![s]\!]_M$; $[\![t \dagger s]\!]_M := [\![t]\!]_M \dagger [\![s]\!]_M$; $[\![t^-]\!]_M := |M|^2 \setminus [\![t]\!]_M$. Here, $\triangle(X)$ denotes the *diagonal relation* (i.e., $\{\langle v, w\rangle \in X^2 \mid v = w\}$). We also define the binary relation $\models \subseteq \mathbf{M}_2 \times \mathrm{Term}^{\mathrm{CoR}}$ by $\langle M, v, w\rangle \models t$ iff $\langle v, w\rangle \in [\![t]\!]_M$.

Proposition 2.1. *The following hold.*

- $\langle M, v_1, v_2\rangle \models t^\pi \iff \langle M, v_{\pi(1)}, v_{\pi(2)}\rangle \models t$.
- $\langle M, v, w\rangle \models a \iff \langle v, w\rangle \in a^M$.
- $\langle M, v, w\rangle \models \mathbf{1} \iff v = w$.
- $\tilde{M} \models \top$ *is always true.*
- $\tilde{M} \models t \cap s \iff \tilde{M} \models t$ *and* $\tilde{M} \models s$.
- $\langle M, v, v'\rangle \models t \cdot s \iff$ *for some* w, $\langle M, v, w\rangle \models t$ *and* $\langle M, w, v'\rangle \models s$.
- $\tilde{M} \models \mathbf{0}$ *is always false.*
- $\tilde{M} \models t \cup s \iff \tilde{M} \models t$ *or* $\tilde{M} \models s$.
- $\langle M, v, v'\rangle \models t \dagger s \iff$ *for every* w, $\langle M, v, w\rangle \models t$ *or* $\langle M, w, v'\rangle \models s$.
- $\tilde{M} \models t^- \iff$ *not* $\tilde{M} \models t$.

Remark 2.2 *(projection and converse).* We only need to define t^π for the case of $\pi = \{1 \mapsto 2, 2 \mapsto 1\}$ (i.e., the case when t^π denotes the converse of t), because in the other cases, t^π can be expressed by not using the π as follows: $t^{\{1\mapsto 1,2\mapsto 2\}} = t$, $t^{\{1\mapsto 1,2\mapsto 1\}} = (t \cap \mathbf{1}) \cdot \top$, and $t^{\{1\mapsto 2,2\mapsto 2\}} = \top \cdot (t \cap \mathbf{1})$. Nevertheless, we introduce t^π for each function $\pi : [2] \to [2]$ for clarifying the relationship between dot-dagger alternation hierarchies in CoR and quantifier alternation hierarchies in FO^3 (see Section 3.1).

Finally, the *size*[1] $\|t\|$ of a CoR term t is defined by: $\|a\| := \|\mathbf{1}\| := \|\top\| := \|\mathbf{0}\| := 1$, $\|t \cup s\| := \|t \cap s\| := \|t \cdot s\| := \|t \dagger s\| := 1 + \|t\| + \|s\|$, and $\|t^-\| := \|t^\pi\| := 1 + \|t\|$.

First-order logic and its fragments

We introduce first-order logic (FO) and its syntactic fragments (see, e.g., [27]): *existential positive logic* (EP) and *primitive positive logic* (PP). Let $\mathscr{V}$ be a countably infinite set of *(first-order) variables*. We use x, y, z, u or w to denote these variables. The set of *formulas* of FO/EP/PP is given by the following grammar, where $a \in \mathscr{A}$ and $x, y \in \mathscr{V}$:

$$\varphi, \psi \in \mathrm{Fml}^{\mathrm{FO}} ::= a(x, y) \mid x = y \mid \mathsf{tt} \mid \varphi \wedge \psi \mid \exists x.\varphi \mid \mathsf{ff} \mid \varphi \vee \psi \mid \forall x.\varphi \mid \neg\varphi$$

$$\varphi, \psi \in \mathrm{Fml}^{\mathrm{EP}} ::= a(x, y) \mid x = y \mid \mathsf{tt} \mid \varphi \wedge \psi \mid \exists x.\varphi \mid \mathsf{ff} \mid \varphi \vee \psi$$

$$\varphi, \psi \in \mathrm{Fml}^{\mathrm{PP}} ::= a(x, y) \mid x = y \mid \mathsf{tt} \mid \varphi \wedge \psi \mid \exists x.\varphi$$

We often use parentheses in ambiguous situations. For a finite sequence $\langle\varphi_i\rangle_{i\in[n]}$ of formulas, we write $\bigvee_{i\in[n]}\varphi_i$ for $\varphi_1 \vee \cdots \vee \varphi_n$ if $n > 0$ and for ff if $n = 0$; and we write $\bigwedge_{i\in[n]}\varphi_i$ for $\varphi_1 \wedge \cdots \wedge \varphi_n$ if $n > 0$ and for tt if $n = 0$. For the sake of brevity, we may identify formulas modulo the commutative and associative laws of $\vee$ and $\wedge$. For a finite set $\Phi = \{\varphi_1, \ldots, \varphi_n\}$ of cardinality n, we write $\bigvee\Phi$ for $\bigvee_{i\in[n]}\varphi_i$, and $\bigwedge\Phi$ for $\bigwedge_{i\in[n]}\varphi_i$.

The *semantics* $\langle M, I\rangle \models \varphi$ of an FO formula φ on M and $I : \mathscr{V} \to |M|$ is a truth value, which is defined in a standard way as follows: $\langle M, I\rangle \models a(x, y) :\Leftrightarrow \langle I(x), I(y)\rangle \in a^M$; $\langle M, I\rangle \models x = y :\Leftrightarrow I(x) = I(y)$; $\langle M, I\rangle \models \mathsf{tt} :\Leftrightarrow$ true; $\langle M, I\rangle \models \mathsf{ff} :\Leftrightarrow$ false; $\langle M, I\rangle \models \varphi \vee \psi :\Leftrightarrow \langle M, I\rangle \models \varphi$ or $\langle M, I\rangle \models \psi$; $\langle M, I\rangle \models \varphi \wedge \psi :\Leftrightarrow \langle M, I\rangle \models \varphi$ and $\langle M, I\rangle \models \psi$; $\langle M, I\rangle \models \neg\varphi :\Leftrightarrow$ not $\langle M, I\rangle \models \varphi$; $\langle M, I\rangle \models \exists x.\varphi :\Leftrightarrow$ for some v, $\langle M, I[v/x]\rangle \models \varphi$; and $\langle M, I\rangle \models \forall x.\varphi :\Leftrightarrow$ for every v, $\langle M, I[v/x]\rangle \models \varphi$. Here, $I[v/x]$ denotes the variable interpretation which maps x to v and agrees with I on all other variables.

$\mathbf{FV}(\varphi)$ denotes the set of *free variables* occurring in φ and $\mathbf{V}(\varphi)$ denotes the set of variables occurring in φ, respectively. For each $\mathrm{Fml}^{\mathcal{C}}$, the fragments $\mathrm{Fml}^{\mathcal{C}^k}$ and $\mathrm{Fml}^{\mathcal{C}^{k(l)}}$ are defined by: $\mathrm{Fml}^{\mathcal{C}^k} := \{\varphi \in \mathrm{Fml}^{\mathcal{C}} \mid \#(\mathbf{V}(\varphi)) \le k\}$ and $\mathrm{Fml}^{\mathcal{C}^{k(l)}} := \{\varphi \in \mathrm{Fml}^{\mathcal{C}^k} \mid \#(\mathbf{FV}(\varphi)) \le l\}$. Also, we define the set of *$\mathcal{C}$(binary-relation-)terms* as follows, where x and y are distinct and φ is a $\mathcal{C}$ formula with $\mathbf{FV}(\varphi) \subseteq \{x, y\}$:

$$t, s \in \mathrm{Term}^{\mathcal{C}} ::= [\varphi]_{x,y}.$$

The *semantics* $[\![[\varphi]_{x,y}]\!]_{M,I}$ is defined as follows:

$$[\![[\varphi]_{x,y}]\!]_{M,I} := \{\langle v, w\rangle \in |M|^2 \mid I[v/x][w/y] \models \varphi\}.$$

We always denote $[\![[\varphi]_{x,y}]\!]_{M,I}$ by $[\![[\varphi]_{x,y}]\!]_M$, because I is irrelevant by the definition of terms (note that $\mathbf{FV}(\varphi) \subseteq \{x, y\}$). We also define the binary relation $\models \subseteq \mathbf{M}_2 \times \mathrm{Term}^{\mathcal{C}}$ by: $\langle M, v, w\rangle \models [\varphi]_{x,y}$ iff $\langle v, w\rangle \in [\![[\varphi]_{x,y}]\!]_M$.

Remark 2.3 *(existential positive logics and conjunctive queries).* The class of PP (resp. EP) formulas in prenex normal form is also known as the class of *conjunctive queries* (resp. *conjunctive queries with union*), which is a major class in database theory (see, e.g., [28, Sect. 4]). However, in this paper, we do not use the prenex normal form, because we are interested in the number of variables of formulas.

Finally, the *size* $\|\varphi\|$ of an FO formula φ is defined by: $\|a(x, y)\| := \|x = y\| := \|\mathsf{tt}\| := \|\mathsf{ff}\| := 1$, $\|\varphi \vee \psi\| := \|\varphi \wedge \psi\| = 1 + \|\varphi\| + \|\psi\|$, and $\|\neg\varphi\| := \|\exists x.\varphi\| := \|\forall x.\varphi\| = 1 + \|\varphi\|$. Also, the *size* $\|[\varphi]_{x,y}\|$ of an FO term $[\varphi]_{x,y}$ is defined as $\|\varphi\|$.

Expressive power and succinctness

We say that two formulas, φ and ψ, are *(semantically) equivalent*, written $\varphi =_{\mathbf{M}} \psi$ if $\langle M, I\rangle \models \varphi \Leftrightarrow \langle M, I\rangle \models \psi$ for every $\langle M, I\rangle$. We say that two (binary-relation-)terms, t and s, are *(semantically) equivalent*, written $t =_{\mathbf{M}_2} s$ if $\tilde{M} \models t \Leftrightarrow \tilde{M} \models s$ for every 2-pointed structure $\tilde{M}$ in $\mathbf{M}_2$ (i.e., $[\![t]\!]_M = [\![s]\!]_M$ for every structure M in $\mathbf{M}$).

[1] Note that many different notions of *size* exist. The size defined here (so-called, *tree-size*) is close to the length of the formula as a string. In Section 6.3, we also consider *DAG-size*, which is close to the number of subformulas of the formula.

Definition 2.4 *(expressive power).* For sets $\mathfrak{T}, \mathfrak{S}$ of terms, we say that $\mathsf{T}\colon \mathfrak{T} \to \mathfrak{S}$ is a *translation* from $\mathfrak{T}$ to $\mathfrak{S}$ if for every term $t \in \mathfrak{T}$, $t =_{\mathbf{M}_2} \mathsf{T}(t)$. We say that $\mathfrak{S}$ is *at least as expressive as* $\mathfrak{T}$, written $\mathfrak{T} \preceq_{\mathbf{M}_2} \mathfrak{S}$ if there exists some translation from $\mathfrak{T}$ to $\mathfrak{S}$. We say that $\mathfrak{T}$ has *the same expressive power as* $\mathfrak{S}$, written $\mathfrak{T} =_{\mathbf{M}_2} \mathfrak{S}$ if both $\mathfrak{T} \preceq_{\mathbf{M}_2} \mathfrak{S}$ and $\mathfrak{S} \preceq_{\mathbf{M}_2} \mathfrak{T}$ hold. We write $\mathfrak{T} \precneqq_{\mathbf{M}_2} \mathfrak{S}$ if both $\mathfrak{T} \preceq_{\mathbf{M}_2} \mathfrak{S}$ and $\mathfrak{S} \not\preceq_{\mathbf{M}_2} \mathfrak{T}$ hold.

Also, for succinctness, we generalize Definition 2.4 as follows.

Definition 2.5 *(succinctness, cf. [12]).* Let $f\colon \mathbb{N}_+ \to \mathbb{N}_+$ be a function. For sets $\mathfrak{T}, \mathfrak{S}$ of terms, we say that $\mathsf{T}\colon \mathfrak{T} \to \mathfrak{S}$ is an *f-size translation* from $\mathfrak{T}$ to $\mathfrak{S}$ if for every term $t \in \mathfrak{T}$, $t =_{\mathbf{M}_2} \mathsf{T}(t)$ and $\|\mathsf{T}(t)\| \leq f(\|t\|)$. We say that $\mathfrak{T}$ is *f-succinct* in $\mathfrak{S}$, written $\mathfrak{T} \preceq^{f}_{\mathbf{M}_2} \mathfrak{S}$ if there exists some f-size translation from $\mathfrak{T}$ to $\mathfrak{S}$. We write $\mathfrak{T} =^{f}_{\mathbf{M}_2} \mathfrak{S}$ if both $\mathfrak{T} \preceq^{f}_{\mathbf{M}_2} \mathfrak{S}$ and $\mathfrak{S} \preceq^{f}_{\mathbf{M}_2} \mathfrak{T}$ hold.

Similarly, let $\mathcal{F} \subseteq \mathbb{N}_+^{\mathbb{N}_+}$ be a function class. For sets $\mathfrak{T}, \mathfrak{S}$ of terms, we say that $\mathsf{T}\colon \mathfrak{T} \to \mathfrak{S}$ is an *$\mathcal{F}$-size translation* from $\mathfrak{T}$ to $\mathfrak{S}$ if T is an f-size translation for some $f \in \mathcal{F}$. We say that $\mathfrak{T}$ is *$\mathcal{F}$-succinct* in $\mathfrak{S}$, written $\mathfrak{T} \preceq^{\mathcal{F}}_{\mathbf{M}_2} \mathfrak{S}$ if there exists some $\mathcal{F}$-size translation from $\mathfrak{T}$ to $\mathfrak{S}$. We write $\mathfrak{T} =^{\mathcal{F}}_{\mathbf{M}_2} \mathfrak{S}$ if both $\mathfrak{T} \preceq^{\mathcal{F}}_{\mathbf{M}_2} \mathfrak{S}$ and $\mathfrak{S} \preceq^{\mathcal{F}}_{\mathbf{M}_2} \mathfrak{T}$ hold.

In this paper, for simplicity, we may write $\mathcal{C}$ for $\mathrm{Term}^{\mathcal{C}}$ (e.g., $\mathrm{PCoR} \preceq^{\mathcal{F}}_{\mathbf{M}_2} \mathrm{EP}^3$ denotes $\mathrm{Term}^{\mathrm{PCoR}} \preceq^{\mathcal{F}}_{\mathbf{M}_2} \mathrm{Term}^{\mathrm{EP}^3}$).

We also consider composing two translations (Proposition 2.7). For two function classes $\mathcal{F}$ and $\mathcal{G}$, we use $\mathcal{F} \circ \mathcal{G}$ to denote the *composition* of $\mathcal{F}$ and $\mathcal{G}$ defined by $\{f \circ g \mid f \in \mathcal{F}, g \in \mathcal{G}\}$.

Definition 2.6. Let $f, g \in \mathbb{N}_+^{\mathbb{N}_+}$ be functions. We say that f is *bounded by* g, written $f \leq_{\mathrm{b}} g$ if $f(n) \leq g(n)$ for every $n \in \mathbb{N}_+$. Let $\mathcal{F}, \mathcal{G} \subseteq \mathbb{N}_+^{\mathbb{N}_+}$ be function classes. We say that $\mathcal{F}$ is *bounded by* $\mathcal{G}$, written $\mathcal{F} \leq_{\mathrm{b}} \mathcal{G}$ if for every $f \in \mathcal{F}$, there exists some $g \in \mathcal{G}$ such that $f \leq_{\mathrm{b}} g$.

Also, we say that a function $f\colon \mathbb{N}_+ \to \mathbb{N}_+$ is *monotone* if for every $n, m \in \mathbb{N}_+$, $n \leq m \Longrightarrow f(n) \leq f(m)$. Then, the following holds (the classes we are interested in are always bounded by monotone functions).

Proposition 2.7. *If $\mathcal{G}$ is bounded by monotone functions (of $\mathcal{G}$), then* $(\preceq^{\mathcal{F}}_{\mathbf{M}_2}) \cdot (\preceq^{\mathcal{G}}_{\mathbf{M}_2}) \subseteq (\preceq^{\mathcal{G}\circ\mathcal{F}}_{\mathbf{M}_2})$.

Proof. Let $\mathfrak{T} \preceq^{\mathcal{F}}_{\mathbf{M}_2} \mathfrak{S} \preceq^{\mathcal{G}}_{\mathbf{M}_2} \mathfrak{U}$. Let $g \in \mathcal{G}$ be a monotone function such that for every $s \in \mathfrak{S}$, there exists some $u \in \mathfrak{U}$ such that $\|u\| \leq g(\|s\|)$. Let f be such that for every $t \in \mathfrak{T}$, there exists some $s \in \mathfrak{S}$ such that $\|s\| \leq f(\|t\|)$. Then, $g(\|s\|) \leq g(f(\|t\|))$ since g is monotone. Combining them, for every $t \in \mathfrak{T}$, there exists some $u \in \mathfrak{U}$ such that $\|u\| \leq (g \circ f)(\|t\|)$. Thus, we obtain a $\mathcal{G} \circ \mathcal{F}$-size translation from $\mathfrak{T}$ to $\mathfrak{U}$ by mapping each t into the u above. Hence, $\mathfrak{T} \preceq^{\mathcal{G}\circ\mathcal{F}}_{\mathbf{M}_2} \mathfrak{U}$. □

For a function $f\colon \mathbb{N}_+ \to \mathbb{N}_+$, we use the following standard asymptotic notations $\mathcal{O}(f), o(f) \subseteq \mathbb{N}_+^{\mathbb{N}_+}$, where $\mathbb{R}_+$ denotes the set of all positive real numbers:

$$\mathcal{O}(f) := \{g \mid \text{there are } c \in \mathbb{R}_+ \text{ and } n_0 \text{ s.t. for all } n \geq n_0,\ g(n) \leq cf(n)\};$$
$$o(f) := \{g \mid \text{for all } c \in \mathbb{R}_+,\ \text{there is } n_0 \text{ s.t. for all } n \geq n_0,\ g(n) < cf(n)\}.$$

We may write $\mathcal{O}(f(n))$ for $\mathcal{O}(f)$ (e.g., $\mathcal{O}(n^2)$ denotes $\mathcal{O}(\lambda n.n^2)$) and write $\mathcal{O}(\mathcal{F})$ for $\bigcup_{f\in\mathcal{F}} \mathcal{O}(f)$. (Similarly for o, we may write $o(f(n))$ for $o(f)$ and write $o(\mathcal{F})$ for $\bigcup_{f\in\mathcal{F}} o(f)$.) Also, we write $f(\mathcal{F})$ for $\{f\} \circ \mathcal{F}$ (e.g., $2^{\mathcal{O}(n)}$ denotes $\{\lambda n.2^n\} \circ \mathcal{O}(n)$).

In particular, we say that the function class $\mathcal{O}(n)$ is *linear*, $\mathcal{O}(n^{\mathcal{O}(1)})$ is *polynomial*, $2^{\mathcal{O}(n)}$ is *exponential*, and $2^{o(n)}$ is *sub-exponential*, respectively.

The standard translation

We recall that from CoR to FO^3, there is an efficient translation [2]; see the $\mathrm{ST}_{x,y}$ in Fig. 1 (where $x \neq y$). Here, $\varphi[y_1y_2/x_1x_2]$ denotes the formula φ in which for each $i \in [2]$, each free variable x_i in φ has been replaced with y_i. It follows that $[\![[\mathrm{ST}_{x,y}(t)]_{x,y}]\!]_M = [\![t]\!]_M$ by simple induction on the structure of t, hence the following theorem. Note that (2) and (3) are also immediate from the standard translation (notice that $\neg$ and $\forall$ do not occur in $\mathrm{ST}_{x,y}(t)$ if $\bullet^-$ and $\dagger$ do not occur in t; furthermore, $\neg$, $\forall$, $\vee$, and ff do not occur in $\mathrm{ST}_{x,y}(t)$ if $\bullet^-$, $\dagger$, $\cup$, and $\mathbf{0}$ do not occur in t).

Theorem 2.8 *([2]).*

(1) $\mathrm{CoR} \preceq^{\mathcal{O}(n)}_{\mathbf{M}_2} \mathrm{FO}^3$.
(2) $\mathrm{PCoR} \preceq^{\mathcal{O}(n)}_{\mathbf{M}_2} \mathrm{EP}^3$.
(3) $\mathrm{PPCoR} \preceq^{\mathcal{O}(n)}_{\mathbf{M}_2} \mathrm{PP}^3$.

$$
\begin{array}{rlrl}
\mathrm{ST}_{x,y}(a) & := a(x,y) & \mathrm{ST}_{x,y}(\top) & := \mathsf{tt} \\
\mathrm{ST}_{x,y}(\mathbf{0}) & := \mathsf{ff} & \mathrm{ST}_{x,y}(\mathbf{1}) & := x = y \\
\mathrm{ST}_{x,y}(t^{-}) & := \neg \mathrm{ST}_{x,y}(t) & \mathrm{ST}_{x_1,x_2}(t^{\pi}) & := \mathrm{ST}_{x_1,x_2}(t)[x_{\pi(1)}x_{\pi(2)}/x_1x_2] \\
\mathrm{ST}_{x,y}(t \cup s) & := \mathrm{ST}_{x,y}(t) \vee \mathrm{ST}_{x,y}(s) & \mathrm{ST}_{x,y}(t \cdot s) & := \exists z.\mathrm{ST}_{x,z}(t) \wedge \mathrm{ST}_{z,y}(s) \\
\mathrm{ST}_{x,y}(t \cap s) & := \mathrm{ST}_{x,y}(t) \wedge \mathrm{ST}_{x,y}(s) & \mathrm{ST}_{x,y}(t \dagger s) & := \forall z.\mathrm{ST}_{x,z}(t) \vee \mathrm{ST}_{z,y}(s)
\end{array}
$$

Fig. 1. The standard translation, where z is a variable such that $z \neq x$ and $z \neq y$.

3. Expressive power equivalence of PCoR and EP^3

In this section, we consider the converse direction of the standard translation, i.e., from FO^3 terms to CoR terms. This section aims to show the following.[2]

Theorem 3.1.

(1) $\mathrm{FO}^3 \leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2} \mathrm{CoR}$.
(2) $\mathrm{EP}^3 \leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2} \mathrm{PCoR}$.
(3) $\mathrm{PP}^3 \leq^{\mathcal{O}(n)}_{\mathbf{M}_2} \mathrm{PPCoR}$.

Combining Theorem 2.8 and 3.1, we obtain the following expressive power equivalences. Note that the expressive power equivalence $\mathrm{CoR} =_{\mathbf{M}_2} \mathrm{FO}^3$ was already shown in [4].

Corollary 3.2.

(1) $\mathrm{CoR} =^{\mathcal{O}(2^n)}_{\mathbf{M}_2} \mathrm{FO}^3$.
(2) $\mathrm{PCoR} =^{\mathcal{O}(2^n)}_{\mathbf{M}_2} \mathrm{EP}^3$.
(3) $\mathrm{PPCoR} =^{\mathcal{O}(n)}_{\mathbf{M}_2} \mathrm{PP}^3$.

From here, we prove Theorem 3.1 by giving a new translation from FO^3 to CoR, which is constructed in the following steps.

(‡1) Translate the given FO^3 term into a term in *negation normal form*.
(‡2) For each sub-formula of the form $\exists z.\psi$ (resp. $\forall z.\psi$), substitute ψ with an equivalent formula, which is a conjunction (resp. disjunction) of formulas having at most two free variables.
(‡3) Push the quantifiers deeper into the formula as much as possible. Then, each sub-formula $\exists z.\varphi$ (resp. $\forall z.\varphi$) is of the form $\exists z.\psi \wedge \rho$ (resp. $\forall z.\psi \vee \rho$) such that $\mathbf{FV}(\psi) \subseteq \{x, z\}$ and $\mathbf{FV}(\rho) \subseteq \{z, y\}$, where x, y, z are three distinct variables.
(‡4) Translate the FO^3 term preprocessed by the above translations to a CoR term by simple structural induction.

In the following, we describe the details of each step.

(‡1): The set of *negation normal form* FO (written NNF-FO, for short) formulas is defined by the following grammar where $a \in \mathscr{A}$:

$$
\begin{aligned}
\varphi, \psi \in \mathrm{Fml}^{\text{NNF-FO}} ::= {} & a(x,y) \mid \bar{a}(x,y) \mid x = y \mid x \neq y \mid \mathsf{tt} \mid \mathsf{ff} \\
& \mid \varphi \vee \psi \mid \varphi \wedge \psi \mid \exists x.\varphi \mid \forall x.\varphi.
\end{aligned}
$$

For an NNF-FO formula φ, we say that φ is a *positive atomic formula* if φ is of the form $a(x, y)$, $x = y$, or tt; and that φ is a *negative atomic formula* if φ is of the form $\bar{a}(x, y)$, $x \neq y$, or ff; and that φ is an *atomic formula* if φ is a positive or negative atomic formula. We may use $\tilde{\varphi}$ to denote an atomic formula. The semantics of NNF-FO is defined by extending the semantics of FO by $\langle M, I\rangle \models \bar{a}(x, y)$ iff $\langle M, I\rangle \models \neg a(x, y)$; and $\langle M, I\rangle \models x \neq y$ iff $\langle M, I\rangle \models \neg(x = y)$, respectively. The size of NNF-FO is defined by extending the size of FO by $\|\bar{a}(x, y)\| := \|x \neq y\| := 1$. Then, the following holds. The proof is straightforward by using De Morgan's law and the double negation elimination law.

Lemma 3.3. *For each k,* $\mathrm{FO}^k \leq^{\lambda n.n}_{\mathbf{M}_2}$ *NNF-*FO^k.

[2] Theorem 3.1 (1) and (2) are improved from the conference version [1] which showed that $\mathrm{FO}^3 \leq^{2^{\mathcal{O}(n)}}_{\mathbf{M}_2} \mathrm{CoR}$ and $\mathrm{EP}^3 \leq^{2^{\mathcal{O}(n)}}_{\mathbf{M}_2} \mathrm{PCoR}$.

Let $\mathsf{T}_\bullet(\psi_l) = \langle \Psi^{(\bullet)}_{l,i} \rangle_{i\in[n_l]}$ for $\bullet \in \{\exists, \forall\}$ and $l \in [2]$.

For $\mathsf{T}_\exists, \mathsf{T}_\forall$:

$$\mathsf{T}_\exists(\tilde{\varphi}) := \mathsf{T}_\forall(\tilde{\varphi}) := \langle\langle \tilde{\varphi} \rangle\rangle$$

$$\mathsf{T}_\exists(\exists z.\psi_1) := \mathsf{T}_\forall(\exists z.\psi_1) := \langle\langle \bigvee_{i\in[n_1]} \exists z. \bigwedge \Phi^{(\exists)}_{1,i} \rangle\rangle$$

$$\mathsf{T}_\exists(\forall z.\psi_1) := \mathsf{T}_\forall(\forall z.\psi_1) := \langle\langle \bigwedge_{i\in[n_1]} \forall z. \bigvee \Phi^{(\forall)}_{1,i} \rangle\rangle$$

$$\mathsf{T}_\exists(\psi_1 \wedge \psi_2) := \langle \Psi^{(\exists)}_{1,i_1} \cdot \Psi^{(\exists)}_{2,i_2} \rangle_{(i_1-1)\times n_2 + i_2 \in [n_1 \times n_2]} \qquad \mathsf{T}_\forall(\psi_1 \wedge \psi_2) := \mathsf{T}_\forall(\psi_1) \cdot \mathsf{T}_\forall(\psi_2)$$

$$\mathsf{T}_\forall(\psi_1 \vee \psi_2) := \langle \Psi^{(\forall)}_{1,i_1} \cdot \Psi^{(\forall)}_{2,i_2} \rangle_{(i_1-1)\times n_2 + i_2 \in [n_1 \times n_2]} \qquad \mathsf{T}_\exists(\psi_1 \vee \psi_2) := \mathsf{T}_\exists(\psi_1) \cdot \mathsf{T}_\exists(\psi_2)$$

For T_-:

$$\mathsf{T}_-(\tilde{\varphi}) := \tilde{\varphi}$$

$$\mathsf{T}_-(\exists z.\psi_1) := \bigvee_{i\in[n_1]} \exists z. \bigwedge \Psi^{(\exists)}_{1,i}$$

$$\mathsf{T}_-(\forall z.\psi_1) := \bigwedge_{i\in[n_1]} \forall z. \bigvee \Psi^{(\forall)}_{1,i}$$

$$\mathsf{T}_-(\psi_1 \wedge \psi_2) := \mathsf{T}_-(\psi_1) \wedge \mathsf{T}_-(\psi_2)$$

$$\mathsf{T}_-(\psi_1 \vee \psi_2) := \mathsf{T}_-(\psi_1) \vee \mathsf{T}_-(\psi_2)$$

Fig. 2. Translation to good-FO^3 formulas.

Remark 3.4. In Lemma 3.3, the definition of the size is particularly important for showing $\leq^{\lambda.n}_{\mathbf{M}_2}$ (i.e., the size of the translated NNF-FO^k formula is at most the size of the original FO^k formula). If we use $\neg$ instead of $\bar{\bullet}$, the translated formula possibly has a larger size than the original formula. For example, $\neg(a(x,y) \wedge b(x,y))$ has size 4. Then, its negation normal form (using $\neg$ instead of $\bar{\bullet}$) is $\neg a(x,y) \vee \neg b(x,y)$ and the formula has size 5, whereas $\|\bar{a}(x,y) \vee \bar{b}(x,y)\| = 3$. This is important to show $\leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}$, rather than $\leq^{2^{\mathcal{O}(n)}}_{\mathbf{M}_2}$.

(‡2): We say that an NNF-FO^3 formula is *good* (written good-FO^3) if, for every sub-formula of the form $\exists z.\psi$ (resp. $\forall z.\psi$), ψ is a conjunction (resp. disjunction) of FO$^{3(2)}$ formulas (i.e., FO^3 formulas with *at most two free variables*). If this condition holds, each sub-formula of the form $\exists z.\psi$ (resp. $\forall z.\psi$) can be written as $\exists z.\rho^{\{x,y\}} \wedge \rho^{\{y,z\}} \wedge \rho^{\{z,x\}}$ (resp. $\forall z.\rho^{\{x,y\}} \vee \rho^{\{y,z\}} \vee \rho^{\{z,x\}}$) by the associativity and commutativity of $\wedge/\vee$, where $\mathbf{FV}(\rho^{\{x,y\}}) \subseteq \{x,y\}$, $\mathbf{FV}(\rho^{\{y,z\}}) \subseteq \{y,z\}$, and $\mathbf{FV}(\rho^{\{z,x\}}) \subseteq \{z,x\}$, and x, y, z are three distinct variables. This property will be useful in the translation (‡3). In this step, we translate NNF-FO^3 terms into good-FO^3 terms. This step is the only step involving an exponential blowup among (‡1)-(‡4).

Lemma 3.5. *NNF-*FO^3 $\leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}$ *good-*FO^3.

To construct a translation for this lemma, in Fig. 2, we mutually define two functions, $\mathsf{T}_\exists, \mathsf{T}_\forall \colon \mathrm{Fml}^{\text{NNF-FO}^3} \to ((\mathrm{Fml}^{\text{good-FO}^{3(2)}})^+)^+$ (i.e., the codomain is the sets of all nonempty finite sequences of nonempty finite sequences of good-FO$^{3(2)}$ formulas), and we define a function $\mathsf{T}_- \colon \mathrm{Fml}^{\text{NNF-FO}^3} \to \mathrm{Fml}^{\text{good-FO}^3}$. Here, $\langle a_1, \ldots, a_{n'}\rangle \cdot \langle a_{n'+1}, \ldots, a_n\rangle$ denotes the sequence $\langle a_1, \ldots, a_n\rangle$. Intuitively, $\mathsf{T}_\exists(\varphi) = \langle\langle \psi_{i,j}\rangle_{j\in[m_i]}\rangle_{i\in[n]}$ (resp. $\mathsf{T}_\forall(\varphi) = \langle\langle \psi_{i,j}\rangle_{j\in[m_i]}\rangle_{i\in[n]}$) denotes the disjunctive normal form $\bigvee_{i\in[n]} \bigwedge_{j\in[m_i]} \psi_{i,j}$ (resp. the conjunctive normal form $\bigwedge_{i\in[n]} \bigvee_{j\in[m_i]} \psi_{i,j}$) of φ where each literal $\psi_{i,j}$ is a good-FO$^{3(2)}$ formula. $\mathsf{T}_-(\varphi)$ is the good-FO$^{3(2)}$ formula obtained by using $\mathsf{T}_\exists$ and $\mathsf{T}_\forall$ in the subformula of the form $\exists z.\psi$ and $\forall z.\psi$ and pushing the quantifier inwards, respectively.

Example 3.6. Let $\psi = (a(x,z) \vee b(z,x)) \wedge c(x,y)$ and let $\varphi = \exists z.\psi$. Note that φ is not a good-FO^3 formula. Then $\mathsf{T}_\exists(\psi) = \langle\langle a(x,z), c(x,y)\rangle, \langle b(z,x), c(x,y)\rangle\rangle$, and thus $\mathsf{T}_-(\varphi) = (\exists z.(a(x,z) \wedge c(x,y))) \vee (\exists z.(b(z,x) \wedge c(x,y)))$. The formula $\mathsf{T}_-(\varphi)$ is a good-FO^3 formula such that $\mathsf{T}_-(\varphi) =_{\mathbf{M}} \varphi$.

Then, the translation T_- satisfies the following Proposition 3.7 and 3.8.

Proposition 3.7. *Let φ be any NNF-*FO^3 *formula. Then, the following holds.*

- *Let* $\mathsf{T}_\exists(\varphi) = \langle \Psi_i \rangle_{i\in[n]}$. *Then,* $\varphi =_{\mathbf{M}} \bigvee_{i\in[n]} \bigwedge \Psi_i$;
- *Let* $\mathsf{T}_\forall(\varphi) = \langle \Psi_i \rangle_{i\in[n]}$. *Then,* $\varphi =_{\mathbf{M}} \bigwedge_{i\in[n]} \bigvee \Psi_i$; *and*
- $\varphi =_{\mathbf{M}} \mathsf{T}_-(\varphi)$.

Proof. By induction on the structure of φ. □

Let $\mathbf{w}\colon ((\mathrm{Fml}^{\text{good-FO}^{3(2)}})^{+})^{+} \to \mathbb{N}$ be the function defined by:

$$\mathbf{w}(\langle\langle\psi_{i,j}\rangle_{j\in[m_i]}\rangle_{i\in[n]}) := n + (\sum_{i\in[n]} m_i) + (\sum_{i\in[n]}\sum_{j\in[m_i]} \|\psi_{i,j}\|).$$

Note that we have $\mathbf{w}(\langle\langle\psi_{i,j}\rangle_{j\in[m_i]}\rangle_{i\in[n]}) - 1 - n = \|\bigvee_{i\in[n]}\bigwedge_{j\in[m_i]}\psi_{i,j}\| = \|\bigwedge_{i\in[n]}\bigvee_{j\in[m_i]}\psi_{i,j}\|$. For the size of the translation T_{-}, we have the following.

Proposition 3.8. *Let φ be any NNF-FO^3 formula. Then, the following holds, respectively.*

(1) *For each $\bullet \in \{\exists, \forall\}$, $\mathbf{w}(\mathsf{T}_{\bullet}(\varphi)) \le 2 \times 2^{\|\varphi\|}$;*
(2) $\|\mathsf{T}_{-}(\varphi)\| \le 2 \times 2^{\|\varphi\|}$.

Proof. First, note that the following hold:

- $\mathbf{w}(\Phi_1 \cdot \Phi_2) \le \mathbf{w}(\Phi_1) + \mathbf{w}(\Phi_2)$;
- $\mathbf{w}(\langle\Psi_i \cdot \Psi'_j\rangle_{(i-1)\times n' + j \in [n\times n']}) \le \mathbf{w}(\langle\Psi_i\rangle_{i\in[n]}) \times \mathbf{w}(\langle\Psi'_j\rangle_{j\in[n']})$.

The first inequality is easy. The second inequality is shown by:

$$\begin{aligned}
&\mathbf{w}(\langle\Psi_i \cdot \Psi'_{i'}\rangle_{(i-1)\times n' + i' \in [n\times n']}) \\
&= n \times n' + \sum_{i\in[n]}\sum_{i'\in[n']}(m_i + m'_{i'}) + \sum_{i\in[n]}\sum_{j\in[m_i]}\sum_{i'\in[n']}\sum_{j'\in[m'_{i'}]} \|\psi_{i,j}\| + \|\psi'_{i',j'}\| \\
&= n \times n' + (\sum_{i\in[n]} m_i)n' + n(\sum_{i'\in[n']} m'_{i'}) + (\sum_{i\in[n]}\sum_{j\in[m_i]}\|\psi_{i,j}\|)(\sum_{i'\in[n']} m'_{i'}) + (\sum_{i\in[n]} m_i)(\sum_{i'\in[n']}\sum_{j'\in[m'_{i'}]}\|\psi_{i',j'}\|) \\
&\le (n + (\sum_{i\in[n]} m_i) + \sum_{i\in[n]}\sum_{j\in[m_i]}\|\psi_{i,j}\|)(n' + (\sum_{i'\in[n']} m'_{i'}) + \sum_{i'\in[n']}\sum_{j'\in[m'_{i'}]}\|\psi_{i',j'}\|) \\
&= \mathbf{w}(\langle\Psi_i\rangle_{i\in[n]}) \times \mathbf{w}(\langle\Psi'_j\rangle_{j\in[n']}).
\end{aligned}$$

(1): By induction on the structure of φ. we only write the case of $\bullet = \exists$.
Case $\tilde{\varphi}$: $\mathbf{w}(\mathsf{T}_{\exists}(\tilde{\varphi})) = 1 + 1 + 1 \le 2^{1+\|\varphi\|}$.
Case $\exists z.\psi$: Let $\mathsf{T}_{\exists}(\psi) = \langle\langle\psi_{i,j}\rangle_{j\in[m_i]}\rangle_{i\in[n]}$. Then,

$$\begin{aligned}
\mathbf{w}(\mathsf{T}_{\exists}(\exists z.\psi)) &= 1 + 1 + \|\bigvee_{i\in[n]} \exists z.\bigwedge_{j\in[m_i]}\psi_{i,j}\| \\
&= 2 + (n-1) + \sum_{i\in[n]}\left(1 + (m_i - 1) + \sum_{j\in[m_i]}\|\psi_{i,j}\|\right) \\
&= 1 + n + (\sum_{i\in[n]} m_i) + (\sum_{i\in[n]}\sum_{j\in[m_i]}\|\psi_{i,j}\|) \\
&= 1 + \mathbf{w}(\mathsf{T}_{\exists}(\psi)) \le 1 + 2^{1+\|\psi\|} \le 2^{1+\|\exists z.\psi\|}
\end{aligned}$$

Case $\forall z.\psi$: In the same way as the case $\exists z.\psi$.
Case $\psi_1 \wedge \psi_2$: Let $\mathsf{T}_{\exists}(\psi_1) = \langle\Psi_i\rangle_{i\in[n]}$ and $\mathsf{T}_{\exists}(\psi_2) = \langle\Psi'_j\rangle_{j\in[n']}$. Then,

$$\begin{aligned}
\mathbf{w}(\mathsf{T}_{\exists}(\psi_1 \wedge \psi_2)) &= \mathbf{w}(\langle\Psi_i \cdot \Psi'_j\rangle_{(i-1)\times n' + j\in[n\times n']}) \\
&\le \mathbf{w}(\langle\Psi_i\rangle_{i\in[n]}) \times \mathbf{w}(\langle\Psi'_j\rangle_{j\in[n']}) = \mathbf{w}(\mathsf{T}_{\exists}(\psi_1)) \times \mathbf{w}(\mathsf{T}_{\exists}(\psi_2)) \\
&\le 2^{1+\|\psi_1\|} \times 2^{1+\|\psi_2\|} = 2^{1+\|\psi_1\wedge\psi_2\|}
\end{aligned}$$

$$
\begin{array}{l}
\mathsf{T}([a(x_i,x_j)]_{x_1,x_2}) := a^{\pi} \\
\mathsf{T}([x_i = x_j]_{x_1,x_2}) := \mathbf{1}^{\pi} \\
\mathsf{T}([\bar{a}(x_i,x_j)]_{x_1,x_2}) := (a^{-})^{\pi} \\
\mathsf{T}([x_i \neq x_j]_{x_1,x_2}) := (\mathbf{1}^{-})^{\pi}
\end{array}
\quad \text{where } \pi = \{1 \mapsto i, 2 \mapsto j\}
$$

$$
\begin{array}{ll}
\mathsf{T}([\mathsf{tt}]_{x,y}) := \top & \mathsf{T}([\varphi^{\{x,y\}} \wedge \psi^{\{x,y\}}]_{x,y}) := \mathsf{T}([\varphi^{\{x,y\}}]_{x,y}) \cap \mathsf{T}([\psi^{\{x,y\}}]_{x,y}) \\
\mathsf{T}([\mathsf{ff}]_{x,y}) := \mathbf{0} & \mathsf{T}([\varphi^{\{x,y\}} \vee \psi^{\{x,y\}}]_{x,y}) := \mathsf{T}([\varphi^{\{x,y\}}]_{x,y}) \cup \mathsf{T}([\psi^{\{x,y\}}]_{x,y}) \\
\multicolumn{2}{c}{\mathsf{T}([\exists z.\varphi^{\{x,z\}} \wedge \psi^{\{z,y\}}]_{x,y}) := \mathsf{T}([\varphi^{\{x,z\}}]_{x,z}) \cdot \mathsf{T}([\psi^{\{z,y\}}]_{z,y})} \\
\multicolumn{2}{c}{\mathsf{T}([\forall z.\varphi^{\{x,z\}} \vee \psi^{\{z,y\}}]_{x,y}) := \mathsf{T}([\varphi^{\{x,z\}}]_{x,z}) \dagger \mathsf{T}([\psi^{\{z,y\}}]_{z,y})}
\end{array}
$$

Fig. 3. Translation to CoR terms.

Case $\psi_1 \vee \psi_2$: Then,

$$
\begin{aligned}
\mathbf{w}(\mathsf{T}_{\exists}(\psi_1 \vee \psi_2)) &= \mathbf{w}(\mathsf{T}_{\exists}(\psi_1) \cdot \mathsf{T}_{\exists}(\psi_2)) \\
&\leq \mathbf{w}(\mathsf{T}_{\exists}(\psi_1)) + \mathbf{w}(\mathsf{T}_{\exists}(\psi_2)) \\
&\leq 2^{1+\|\psi_1\|} + 2^{1+\|\psi_2\|} \leq 2^{1+\max(1+\|\psi_1\|,1+\|\psi_2\|)} \leq 2^{1+\|\psi_1 \vee \psi_2\|}
\end{aligned}
$$

(2): By induction on the structure of φ using (1). □

We are now ready to prove Lemma 3.5.

Proof of Lemma 3.5. Let us recall the translation T_{-} in Fig. 2. By Proposition 3.7 and 3.8, the statement of the theorem has been proved using T_{-}. □

(‡3): In this step, we translate good-FO^3 terms into FO^3 terms in the following normal form.

Definition 3.9. For two distinct variables x and y, we say that an FO^3 formula is *($\{x, y\}$-)nice* if it is in the set defined by the following grammar, where $\tilde{x}, \tilde{y} \in \{x, y\}$ and z is the variable distinct from x and y:

$$
\begin{aligned}
\varphi^{\{x,y\}}, \psi^{\{x,y\}} ::= {} & a(\tilde{x},\tilde{y}) \mid \bar{a}(\tilde{x},\tilde{y}) \mid \tilde{x} = \tilde{y} \mid \tilde{x} \neq \tilde{y} \mid \mathsf{tt} \mid \mathsf{ff} \mid \varphi^{\{x,y\}} \vee \psi^{\{x,y\}} \\
& \mid \varphi^{\{x,y\}} \wedge \psi^{\{x,y\}} \mid \exists z.\varphi^{\{x,z\}} \wedge \psi^{\{z,y\}} \mid \forall z.\varphi^{\{x,z\}} \vee \psi^{\{z,y\}}.
\end{aligned}
$$

Intuitively, if an FO^3 term is nice, then it is 'almost' a two-variable term, in that, even if a subformula of the term has three free variables, it should be of the form $\varphi^{\{x,z\}} \wedge \psi^{\{z,y\}}$ or $\varphi^{\{x,z\}} \vee \psi^{\{z,y\}}$; hence its immediate subformulas have at most two free variables.

Lemma 3.10. *good*-FO^3 $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ *nice*-FO^3.

Proof. Let $\mathsf{T}\colon \mathrm{Fml}^{\text{good-FO}^3} \to \mathrm{Fml}^{\text{nice-FO}^3}$ be the translation defined as follows: $\mathsf{T}(\varphi) := \varphi$ if φ is an atomic formula; $\mathsf{T}(\psi \vee \rho) := \mathsf{T}(\psi) \vee \mathsf{T}(\rho)$; $\mathsf{T}(\psi \wedge \rho) := \mathsf{T}(\psi) \wedge \mathsf{T}(\rho)$; $\mathsf{T}(\exists z.\psi) := \mathsf{T}(\psi_1^{\{x,y\}}) \wedge \exists z.\mathsf{T}(\psi_2^{\{x,z\}}) \wedge \mathsf{T}(\psi_3^{\{y,z\}})$ where $\psi =_{\mathbf{M}} \psi_1^{\{x,y\}} \wedge \psi_2^{\{x,z\}} \wedge \psi_3^{\{y,z\}}$ by reordering and using tt appropriately (since $\exists z.\psi$ is a good FO^3 formula); and $\mathsf{T}(\forall z.\psi) := \mathsf{T}(\psi_1^{\{x,y\}}) \vee \forall z.\mathsf{T}(\psi_2^{\{x,z\}}) \vee \mathsf{T}(\psi_3^{\{y,z\}})$ where $\psi =_{\mathbf{M}} \psi_1^{\{x,y\}} \vee \psi_2^{\{x,z\}} \vee \psi_3^{\{y,z\}}$ by reordering and using ff appropriately (since $\forall z.\psi$ is a good FO^3 formula), where each φ^S satisfies $\mathbf{FV}(\varphi^S) \subseteq S$. By straightforward induction on the size of φ, (1) $\mathsf{T}(\varphi) =_{\mathbf{M}} \varphi$ and (2) $\|\mathsf{T}(\varphi)\| \leq 3\|\varphi\|$ (note $\|\mathsf{T}(\exists z.\mathsf{tt})\| = \|\mathsf{tt} \wedge \exists z.\mathsf{tt} \wedge \mathsf{tt}\| = 6$). Hence by using T, the statement of the theorem has been proved. □

(‡4): Finally, we translate nice-FO^3 terms into CoR terms by simple structural induction as follows (Fig. 3).

Lemma 3.11. *nice*-FO^3 $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ CoR.

Proof. By induction on the structure of φ, we can show that (1) for every $\{x, y\}$-nice FO^3 term $[\varphi]_{x,y}$, $\mathsf{T}([\varphi]_{x,y}) =_{\mathbf{M}_2} [\varphi]_{x,y}$; and (2) $\|\mathsf{T}([\varphi]_{x,y})\| \leq 3\|[\varphi]_{x,y}\|$ (note $\|\mathsf{T}([\bar{a}(x_i,x_j)]_{x_1,x_2})\| = 3$). Hence by using T, the statement of the theorem has been proved. □

Combining the aforementioned lemmas, we have:

$$
\text{FO}^3 \leq^{\lambda n.n}_{\mathbf{M}_2} \text{NNF-FO}^3 \leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2} \text{good-FO}^3 \leq^{\mathcal{O}(n)}_{\mathbf{M}_2} \text{nice-FO}^3 \leq^{\mathcal{O}(n)}_{\mathbf{M}_2} \text{CoR}.
$$

Furthermore, by using (‡1)-(‡4), Theorem 3.1 is proved as follows.

Proof of Theorem 3.1. (1): By Proposition 2.7, $(\leq_{\mathbf{M}_2}^{\lambda n.n}) \cdot (\leq_{\mathbf{M}_2}^{\mathcal{O}(2^n)}) \cdot (\leq_{\mathbf{M}_2}^{\mathcal{O}(n)}) \cdot (\leq_{\mathbf{M}_2}^{\mathcal{O}(n)}) \subseteq (\leq_{\mathbf{M}_2}^{\mathcal{O}(n)\circ\mathcal{O}(n)\circ\mathcal{O}(2^n)\circ\{\lambda n.n\}}) = (\leq_{\mathbf{M}_2}^{\mathcal{O}(2^n)})$. Therefore, by combining (‡1)-(‡4), $\mathrm{FO}^{3(2)} \leq_{\mathbf{M}_2}^{\mathcal{O}(2^n)}$ CoR. (2): Since each EP^3 formula does not not contain $\neg$ nor $\forall$, the term translated by (‡1)-(‡4) (more precisely, (‡2)-(‡4) are sufficient) does not contain $\bullet^-$ nor †, so it is a PCoR term. Thus, $\mathrm{EP}^3 \leq_{\mathbf{M}_2}^{\mathcal{O}(2^n)}$ PCoR. (3): Since each PP^3 formula does not not contain $\neg$, $\forall$, $\vee$ nor ff, the term translated by (‡1)-(‡4) does not contain $\bullet^-$, †, $\cup$, nor $\mathbf{0}$, so it is a PPCoR term. In this case, we do not have to use ((‡1) and) (‡2) because every PP^3 formula is always good. Therefore by $(\leq_{\mathbf{M}_2}^{\lambda n.n}) \cdot (\leq_{\mathbf{M}_2}^{\mathcal{O}(n)}) \cdot (\leq_{\mathbf{M}_2}^{\mathcal{O}(n)}) \subseteq (\leq_{\mathbf{M}_2}^{\mathcal{O}(n)\circ\mathcal{O}(n)\circ\{\lambda n.n\}}) = (\leq_{\mathbf{M}_2}^{\mathcal{O}(n)})$, $\mathrm{PP}^3 \leq_{\mathbf{M}_2}^{\mathcal{O}(n)}$ PPCoR. □

3.1. Quantifier alternation and dot-dagger alternation hierarchies

In this subsection, we give a more fine-grained expressive power equivalence between CoR and FO^3 in terms of *the quantifier alternation hierarchy*.

Definition 3.12 *(quantifier alternation hierarchy).* The sets $\{\Sigma_n, \Pi_n\}_{n\in\mathbb{N}}$ are the minimal sets of FO formulas satisfying the following.

- If an FO formula φ contains neither $\exists$ nor $\forall$, then $\varphi \in \Sigma_0$ and $\varphi \in \Pi_0$.
- For $n \geq 0$, $\Sigma_n \cup \Pi_n \subseteq \Sigma_{n+1} \cap \Pi_{n+1}$.
- For $n \geq 1$, if $\varphi, \psi \in \Sigma_n$, then $\varphi \vee \psi, \varphi \wedge \psi, \exists x.\varphi \in \Sigma_n$ and $\forall x.\varphi \in \Pi_{n+1}$.
- For $n \geq 1$, if $\varphi, \psi \in \Pi_n$, then $\varphi \vee \psi, \varphi \wedge \psi, \forall x.\varphi \in \Pi_n$ and $\exists x.\varphi \in \Sigma_{n+1}$.

We also define the sets $\{\Sigma_n^k, \Pi_n^k\}_{n\in\mathbb{N}}$ as the subclasses of FO^k formulas defined by the same rules.

We now define *dot-dagger alternation hierarchy* in CoR in the same manner as the quantifier alternation hierarchy, as follows.

Definition 3.13 *(dot-dagger alternation hierarchy).* The subclasses of CoR terms, $\{\Sigma_n^{\mathrm{CoR}}, \Pi_n^{\mathrm{CoR}}\}_{n\in\mathbb{N}}$, are the minimal sets satisfying the following.

- If a CoR term t contains neither $\cdot$ nor †, then $t \in \Sigma_0^{\mathrm{CoR}}$ and $t \in \Pi_0^{\mathrm{CoR}}$.
- For $n \geq 0$, $\Sigma_n^{\mathrm{CoR}} \cup \Pi_n^{\mathrm{CoR}} \subseteq \Sigma_{n+1}^{\mathrm{CoR}} \cap \Pi_{n+1}^{\mathrm{CoR}}$.
- For $n \geq 1$, if $t, u \in \Sigma_n^{\mathrm{CoR}}$, then t^π, $t \cup u, t \cap u, t \cdot u \in \Sigma_n^{\mathrm{CoR}}$ and $t \dagger u \in \Pi_{n+1}^{\mathrm{CoR}}$.
- For $n \geq 1$, if $t, u \in \Pi_n^{\mathrm{CoR}}$, then t^π, $t \cup u, t \cap u, t \dagger u \in \Pi_n^{\mathrm{CoR}}$ and $t \cdot u \in \Sigma_{n+1}^{\mathrm{CoR}}$.

The following shows that the dot-dagger alternation hierarchy in CoR is expressive power equivalent to the quantifier alternation hierarchy in FO^3, uniformly.

Corollary 3.14. *For each $n \geq 0$, $\Sigma_n^{\mathrm{CoR}} =_{\mathbf{M}_2} \Sigma_n^3$ and $\Pi_n^{\mathrm{CoR}} =_{\mathbf{M}_2} \Pi_n^3$. More precisely, we have:*

- *$\Sigma_n^{\mathrm{CoR}} \leq_{\mathbf{M}_2}^{\mathcal{O}(n)} \Sigma_n^3$ and $\Pi_n^{\mathrm{CoR}} \leq_{\mathbf{M}_2}^{\mathcal{O}(n)} \Pi_n^3$; and*
- *$\Sigma_n^3 \leq_{\mathbf{M}_2}^{\mathcal{O}(2^n)} \Sigma_n^{\mathrm{CoR}}$ and $\Pi_n^3 \leq_{\mathbf{M}_2}^{\mathcal{O}(2^n)} \Pi_n^{\mathrm{CoR}}$.*

Proof sketch. In the standard translation, if a given CoR term is in Σ_n^{CoR} (resp. Π_n^{CoR}), then the translated term is in Σ_n^3 (resp. Π_n^3). Conversely, for each of (‡1)-(‡3), if a given $\mathrm{FO}^{3(2)}$ formula is in Σ_n^3 (resp. Π_n^3), then the translated nice $\mathrm{FO}^{3(2)}$ formula is also in Σ_n^3 (resp. Π_n^3). Also for (‡4), if a given nice FO^3 term is in Σ_n^3 (resp. Π_n^3), then the translated CoR term is in Σ_n^{CoR} (resp. Π_n^{CoR}). (All the above are shown by simple induction on the size of the given term/formula.) Hence, by using the exact translation as the proof of Theorem 3.1, the statement of the theorem has been proved. □

The dot-dagger alternation hierarchy in CoR is also strict as the quantifier alternation hierarchy in FO^3. The strictness of the quantifier alternation hierarchy in FO^3 is already shown in [29, Lemma 3.9] because the Σ_{n+1}-formula constructed in [29, Lemma 3.9] can be trivially expressed by an Σ_{n+1}^2-formula.

Proposition 3.15 *(Cor. of [29, Lemma 3.9]). For each $n \geq 0$, $\Sigma_{n+1}^2 \not\leq_{\mathbf{M}_2} \Sigma_n$.*

Hence, the following holds by $\Sigma_{n+1}^3 \not\leq_{\mathbf{M}_2} \Sigma_n^3$ (Proposition 3.15) and Corollary 3.14.

Corollary 3.16. *For each $n \geq 0$, $\Sigma_{n+1}^{\mathrm{CoR}} \not\leq_{\mathbf{M}_2} \Sigma_n^{\mathrm{CoR}}$. Hence, $\Sigma_n^{\mathrm{CoR}} \lneq_{\mathbf{M}_2} \Sigma_{n+1}^{\mathrm{CoR}}$.*

Additionally, we remark that the following two hold for PCoR.

Proposition 3.17. $\Sigma_0^{\mathrm{CoR}} \not\leq_{\mathbf{M}_2} \mathrm{PCoR}$. *Hence,* $\mathrm{PCoR} \lneq_{\mathbf{M}_2} \Sigma_1^{\mathrm{CoR}}$.

Proof. For every PCoR term t, for every 2-pointed structures $\tilde{M}$ and $\tilde{N}$, if $\tilde{M} \to \tilde{N}$ and $\tilde{M} \models t$, then $\tilde{N} \models t$ (see, e.g., [8]). Let us consider the Σ_0^{CoR} term $t_0 := a^-$. Let $\tilde{M}_\emptyset := \langle \{1,2\}, \{\emptyset\}_{a' \in \mathscr{A}}, 1, 2\rangle$ and $\tilde{N}_a := \langle \{1,2\}, \{R_{a'}\}_{a' \in \mathscr{A}}, 1, 2\rangle$ where $R_{a'} = \{\langle 1,2\rangle\}$ if $a' = a$ and $R_{a'} = \emptyset$ otherwise. Then, $\tilde{M}_\emptyset \to \tilde{N}_a$, and $\tilde{M}_\emptyset \models t_0$, $\tilde{N}_a \not\models t_0$, and thus t_0 does not satisfies the above. Therefore, for every PPCoR term t, $t \neq_{\mathbf{M}_2} t_0$. □

Proposition 3.18. $\mathrm{PCoR} \not\leq_{\mathbf{M}_2} \mathrm{PPCoR}$. *Hence,* $\mathrm{PPCoR} \lneq_{\mathbf{M}_2} \mathrm{PCoR}$.

Proof. For every PPCoR term t, there is a 2-pointed structure $\tilde{M}_t$ such that, for every 2-pointed structure $\tilde{N}$, $\tilde{N} \models t$ iff $\tilde{M}_t \to \tilde{N}$ (see, e.g., [8, Lem. 3]). Let us consider the PCoR term $t_0 := a \cup b$ (where $a \neq b$). Assume that $t_0 =_{\mathbf{M}_2} t$ for some PPCoR term t. Then, there exists some $\tilde{M}_{t_0}$ satisfying the above. First, since $\tilde{M}_{t_0} \to \tilde{M}_{t_0}$, $\tilde{M}_{t_0} \models t_0$. Second, let $\tilde{N}_a := \langle \{1,2\}, \{R_{a'}\}_{a' \in \mathscr{A}}, 1, 2\rangle$ where $R_{a'} = \{\langle 1,2\rangle\}$ if $a' = a$ and $R_{a'} = \emptyset$ otherwise. Similarly for $\tilde{N}_b$. Then, $\tilde{M}_{t_0} \to \tilde{N}_a$ and $\tilde{M}_{t_0} \to \tilde{N}_b$, since $\tilde{N}_a \models t_0$ and $\tilde{N}_b \models t_0$. From them, $R_b^{\tilde{M}_{t_0}} = \emptyset$ and $R_a^{\tilde{M}_{t_0}} = \emptyset$, respectively. Thus, $\tilde{M}_{t_0} \not\models t_0$. However, it reaches a contradiction. Therefore, for every PPCoR term t, $t \neq_{\mathbf{M}_2} t_0$. □

3.2. The calculus of relations for FO^2

In this subsection, we consider the two-variable case. We introduce CoR^2, PCoR^2, and PPCoR^2. The sets of terms are defined by the following grammar, where $a \in \mathscr{A}$.

$$t, s \in \mathrm{Term}^{\mathrm{CoR}^2} ::= t^\pi \mid a \mid \mathbf{1} \mid \top \mid t \cap s \mid t^\exists \mid \mathbf{0} \mid t \cup s \mid t^\forall \mid t^-$$
$$t, s \in \mathrm{Term}^{\mathrm{PCoR}^2} ::= t^\pi \mid a \mid \mathbf{1} \mid \top \mid t \cap s \mid t^\exists \mid \mathbf{0} \mid t \cup s$$
$$t, s \in \mathrm{Term}^{\mathrm{PPCoR}^2} ::= t^\pi \mid a \mid \mathbf{1} \mid \top \mid t \cap s \mid t^\exists$$

The semantics $[\![t]\!]_M$ is defined by extending the definition of the semantics of CoR as follows:

$$[\![t^\exists]\!]_M := \{\langle v, w\rangle \in |M| \mid \text{for some } w' \in |M|, \langle v, w'\rangle \in [\![t]\!]_M\};$$
$$[\![t^\forall]\!]_M := \{\langle v, w\rangle \in |M| \mid \text{for every } w' \in |M|, \langle v, w'\rangle \in [\![t]\!]_M\}.$$

The operation $\bullet^\exists$ is always called *domain* operation. Also, the size $\|t\|$ is defined by extending the definition of the size of CoR by: $\|t^\exists\| := \|t^\forall\| := 1 + \|t\|$. Note that the following holds.

Proposition 3.19.

- $\mathrm{CoR}^2 \leq_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{CoR}$.
- $\mathrm{PCoR}^2 \leq_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{PCoR}$.
- $\mathrm{PPCoR}^2 \leq_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{PPCoR}$.

More precisely, $\mathrm{CoR}^2 \leq_{\mathbf{M}_2}^{\lambda n.3n} \mathrm{CoR}$ *(similarly for the other two inequations).*

Proof. By $t^\exists =_{\mathbf{M}_2} t \cdot \top$ and $t^\forall =_{\mathbf{M}_2} t \dagger \mathbf{0}$. □

Then, the following holds by using the same strategy as in Section 3.

Corollary 3.20.

- $\mathrm{CoR}^2 =_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{FO}^2$.
- $\mathrm{PCoR}^2 =_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{EP}^2$.
- $\mathrm{PPCoR}^2 =_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{PP}^2$.

Proof sketch. The direction $\leq_{\mathbf{M}_2}^{\mathcal{O}(n)}$ is shown by extending the standard translation $\mathrm{ST}_{x,y}$ by: $\mathrm{ST}_{x,y}(t^\exists) := \exists y.\mathrm{ST}_{x,y}(t)$ and $\mathrm{ST}_{x,y}(t^\forall) := \forall y.\mathrm{ST}_{x,y}(t)$. The direction $\geq_{\mathbf{M}_2}^{\mathcal{O}(n)}$ is shown by the translations (‡1) and (‡4). After that (‡1) is applied to FO^2 formulas, they are in the following grammar where $\tilde{x}, \tilde{y} \in \{x, y\}$:

$$\varphi^{\{x,y\}}, \psi^{\{x,y\}} ::= a(\tilde{x}, \tilde{y}) \mid \bar{a}(\tilde{x}, \tilde{y}) \mid \tilde{x} = \tilde{y} \mid \tilde{x} \neq \tilde{y} \mid \mathsf{tt} \mid \mathsf{ff} \mid \varphi^{\{x,y\}} \vee \psi^{\{x,y\}} \mid \varphi^{\{x,y\}} \wedge \psi^{\{x,y\}} \mid \exists \tilde{x}.\varphi^{\{x,y\}} \mid \forall \tilde{x}.\varphi^{\{x,y\}}.$$

(Note that the translations (‡2) and (‡3) are not needed in the two-variable fragment. In particular for (‡2), it is because for each subformula of the form $Q\tilde{x}.\psi$, the formula ψ is itself a conjunction/disjunction of formulas having at most two free variables since ψ is a subformula of a formula having at most two variables.) Then, the translation T in (‡4) works by modifying the definition of T for quantifiers as follows: $\mathsf{T}([Qx.\varphi^{\{x,y\}}]_{x,y}) := \mathsf{T}([\varphi^{\{x,y\}}]_{x,y})^{Q}$ and $\mathsf{T}([Qy.\varphi^{\{x,y\}}]_{x,y}) := \mathsf{T}([\varphi^{\{x,y\}}]_{y,x})^{Q}$ for $Q \in \{\exists, \forall\}$. □

Furthermore, as with Corollary 3.14 and Corollary 3.16, the following holds. Here, $\{\Sigma_n^{\mathrm{CoR}^2}, \Pi_n^{\mathrm{CoR}^2}\}_{n\in\mathbb{N}}$ are defined in the same way as Definition 3.13 (by replacing $\cdot$ with $\bullet^{\exists}$ and $\dagger$ with $\bullet^{\forall}$, respectively).

Corollary 3.21. *The following hold, respectively.*

(1) *For each* $n \geq 0$, $\Sigma_n^{\mathrm{CoR}^2} =_{\mathbf{M}_2}^{\mathcal{O}(n)} \Sigma_n^2$ *and* $\Pi_n^{\mathrm{CoR}^2} =_{\mathbf{M}_2}^{\mathcal{O}(n)} \Pi_n^2$.
(2) *For each* $n \geq 0$, $\Sigma_{n+1}^{\mathrm{CoR}^2} \not\leq_{\mathbf{M}_2} \Sigma_n^{\mathrm{CoR}^2}$.

Proof sketch. (1): In the same manner as Corollary 3.14. (2): By (1) and $\Sigma_{n+1}^2 \not\leq_{\mathbf{M}_2} \Sigma_n^2$ (Proposition 3.15). □

Remark 3.22. By Proposition 3.19 and Corollary 3.20, we have $\mathrm{EP}^2 \leq_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{PCoR}$ (and $\mathrm{FO}^2 \leq_{\mathbf{M}_2}^{\mathcal{O}(n)} \mathrm{CoR}$). We will have $\mathrm{EP}^3 \not\leq_{\mathbf{M}_2}^{2^{o(n)}} \mathrm{PCoR}$ (Theorem 5.1). From them, we see that the exponential blowup only happens in the three-variable case.

Additionally, we remark that the following two hold for PCoR^2.

Proposition 3.23. $\Sigma_0^{\mathrm{CoR}^2} \not\leq_{\mathbf{M}_2} \mathrm{PCoR}^2$. *Hence,* $\mathrm{PCoR}^2 \lneq_{\mathbf{M}_2} \Sigma_1^{\mathrm{CoR}^2}$.

Proof. By the same proof of Proposition 3.17. □

Proposition 3.24. $\mathrm{PCoR}^2 \not\leq_{\mathbf{M}_2} \mathrm{PPCoR}^2$. *Hence,* $\mathrm{PPCoR}^2 \lneq_{\mathbf{M}_2} \mathrm{PCoR}^2$.

Proof. By the same proof of Proposition 3.18. □

4. A formula-size game for CoR

In this section, we introduce a formula-size game for CoR. The game has one player, called Spoiler. The states in the game are all the pairs of sets of 2-pointed structures. We first generalize the satisfaction relation $\models$ for pairs of sets of 2-pointed structures as follows: $\langle \mathcal{S}, \mathcal{U} \rangle \models t$ iff (1) for every $\tilde{M} \in \mathcal{S}$, $\tilde{M} \models t$, and (2) for every $\tilde{M} \in \mathcal{U}$, $\tilde{M} \not\models t$. The following is immediate from the definition.

Proposition 4.1. $\langle \bigcup_{i\in I} \mathcal{S}_i, \bigcup_{i\in I} \mathcal{U}_i \rangle \models t$ *iff* $\langle \mathcal{S}_i, \mathcal{U}_i \rangle \models t$ *for each* $i \in I$.

Also, we define the following operators for sets of 2-pointed structures:

$$\mathcal{C} \cdot \mathcal{D} := \{\langle M, v, v' \rangle \mid \text{for some } w \in |M|,\ \langle M, v, w \rangle \in \mathcal{C} \text{ and } \langle M, w, v' \rangle \in \mathcal{D}\};$$
$$\mathcal{C} \dagger \mathcal{D} := \{\langle M, v, v' \rangle \mid \text{for every } w \in |M|,\ \langle M, v, w \rangle \in \mathcal{C} \text{ or } \langle M, w, v' \rangle \in \mathcal{D}\};$$
$$\mathcal{C}^{\pi} := \{\langle M, v_1, v_2 \rangle \mid \langle M, v_{\pi(1)}, v_{\pi(2)} \rangle \in \mathcal{C}\}.$$

Then, Fig. 4 presents derivation rules for the game. We say that Spoiler *wins* the game $\langle \mathcal{S}, \mathcal{U} \rangle$ if there exists a derivation tree of $\langle \mathcal{S}, \mathcal{U} \rangle \vdash t$ for some t.

We will show the sound- and completeness (Theorem 4.3) of $\vdash$. Essentially, the following lemma implies them.

Lemma 4.2.

($\bullet^{\pi}$) $\langle \mathcal{S}, \mathcal{U} \rangle \models t^{\pi}$ *iff for some* $\mathcal{S}_1, \mathcal{U}_1$ *such that* $\mathcal{S} \subseteq \mathcal{S}_1^{\pi}$ *and* $\mathcal{U} \subseteq \mathcal{U}_1^{\pi}$, $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \models t$.
($\bullet^{-}$) $\langle \mathcal{S}, \mathcal{U} \rangle \models t^{-}$ *iff* $\langle \mathcal{U}, \mathcal{S} \rangle \models t$.
($\cup$) $\langle \mathcal{S}, \mathcal{U} \rangle \models t_1 \cup t_2$ *iff for some* $\mathcal{S}_1, \mathcal{S}_2$ *such that* $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$, $\langle \mathcal{S}_1, \mathcal{U} \rangle \models t_1$ *and* $\langle \mathcal{S}_2, \mathcal{U} \rangle \models t_2$.
($\cap$) $\langle \mathcal{S}, \mathcal{U} \rangle \models t_1 \cap t_2$ *iff for some* $\mathcal{U}_1, \mathcal{U}_2$ *such that* $\mathcal{U} = \mathcal{U}_1 \cup \mathcal{U}_2$, $\langle \mathcal{S}, \mathcal{U}_1 \rangle \models t_1$ *and* $\langle \mathcal{S}, \mathcal{U}_2 \rangle \models t_2$.

$$\frac{\langle \mathcal{S} \cup \mathcal{S}', \mathcal{U} \cup \mathcal{U}' \rangle \vdash t}{\langle \mathcal{S}, \mathcal{U} \rangle \vdash t}\ (\mathrm{WK})$$

$$\frac{(\langle \mathcal{S}, \mathcal{U} \rangle \models \tilde{a})}{\langle \mathcal{S}, \mathcal{U} \rangle \vdash \tilde{a}}\ (\mathrm{At}) \qquad \frac{\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \vdash t}{\langle \mathcal{S}_1^{\pi}, \mathcal{U}_1^{\pi} \rangle \vdash t^{\pi}}\ (\bullet^{\pi}) \qquad \frac{\langle \mathcal{U}, \mathcal{S} \rangle \vdash t}{\langle \mathcal{S}, \mathcal{U} \rangle \vdash t^{-}}\ (\bullet^{-})$$

$$\frac{\langle \mathcal{S}_1, \mathcal{U} \rangle \vdash t_1 \quad \langle \mathcal{S}_2, \mathcal{U} \rangle \vdash t_2}{\langle \mathcal{S}_1 \cup \mathcal{S}_2, \mathcal{U} \rangle \vdash t_1 \cup t_2}\ (\cup) \qquad \frac{\langle \mathcal{S}, \mathcal{U}_1 \rangle \vdash t_1 \quad \langle \mathcal{S}, \mathcal{U}_2 \rangle \vdash t_2}{\langle \mathcal{S}, \mathcal{U}_1 \cup \mathcal{U}_2 \rangle \vdash t_1 \cap t_2}\ (\cap)$$

$$\frac{\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \vdash t_1 \quad \langle \mathcal{S}_2, \mathcal{U}_2 \rangle \vdash t_2}{\langle \mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2 \rangle \vdash t_1 \cdot t_2}\ (\cdot) \qquad \frac{\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \vdash t_1 \quad \langle \mathcal{S}_2, \mathcal{U}_2 \rangle \vdash t_2}{\langle \mathcal{S}_1 \dagger \mathcal{S}_2, \mathcal{U}_1 \cdot \mathcal{U}_2 \rangle \vdash t_1 \dagger t_2}\ (\dagger)$$

Fig. 4. Formula-size game for CoR.

$(\cdot)$ $\langle \mathcal{S}, \mathcal{U} \rangle \models t_1 \cdot t_2$ *iff for some* $\mathcal{S}_1, \mathcal{S}_2, \mathcal{U}_1, \mathcal{U}_2$ *such that* $\mathcal{S} \subseteq \mathcal{S}_1 \cdot \mathcal{S}_2$ *and* $\mathcal{U} \subseteq \mathcal{U}_1 \dagger \mathcal{U}_2$, $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \models t_1$ *and* $\langle \mathcal{S}_2, \mathcal{U}_2 \rangle \models t_2$.

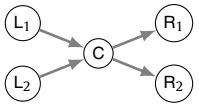
$(\dagger)$ $\langle \mathcal{S}, \mathcal{U} \rangle \models t_1 \dagger t_2$ *iff for some* $\mathcal{S}_1, \mathcal{S}_2, \mathcal{U}_1, \mathcal{U}_2$ *such that* $\mathcal{S} \subseteq \mathcal{S}_1 \dagger \mathcal{S}_2$ *and* $\mathcal{U} \subseteq \mathcal{U}_1 \cdot \mathcal{U}_2$, $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \models t_1$ *and* $\langle \mathcal{S}_2, \mathcal{U}_2 \rangle \models t_2$.

Proof. They are straightforward by the definition of $\models$. We only write the proof of $(\cdot)$. By Proposition 4.1, it suffices to show that (1) $\langle \{\tilde{M}\}, \emptyset \rangle \models t_1 \cdot t_2$ iff for some $\mathcal{S}_1$ and $\mathcal{S}_2$ such that $\tilde{M} \in \mathcal{S}_1 \cdot \mathcal{S}_2$, $\langle \mathcal{S}_1, \emptyset \rangle \models t_1$ and $\langle \mathcal{S}_2, \emptyset \rangle \models t_2$; and (2) $\langle \emptyset, \{\tilde{M}\} \rangle \models t_1 \cdot t_2$ iff for some $\mathcal{U}_1$ and $\mathcal{U}_2$ such that $\tilde{M} \in \mathcal{U}_1 \dagger \mathcal{U}_2$, $\langle \emptyset, \mathcal{U}_1 \rangle \models t_1$ and $\langle \emptyset, \mathcal{U}_2 \rangle \models t_2$. (1)$(\Rightarrow)$: Let $\tilde{M} = \langle M, v, v' \rangle$. By $\tilde{M} \models t_1 \cdot t_2$, there exists w such that $\langle M, v, w \rangle \models t_1$ and $\langle M, w, v' \rangle \models t_2$. Therefore by letting $\mathcal{S}_1 = \{\langle M, v, w \rangle\}$ and $\mathcal{S}_2 = \{\langle M, w, v' \rangle\}$, $\tilde{M} \in \mathcal{S}_1 \cdot \mathcal{S}_2$, $\langle \mathcal{S}_1, \emptyset \rangle \models t_1$ and $\langle \mathcal{S}_2, \emptyset \rangle \models t_2$ hold. (1)$(\Leftarrow)$: By Proposition 4.1, it suffices to show $\langle \mathcal{S}_1 \cdot \mathcal{S}_2, \emptyset \rangle \models t_1 \cdot t_2$. For every $\langle M, v, v' \rangle \in \mathcal{S}_1 \cdot \mathcal{S}_2$, let w be such that $\langle M, v, w \rangle \in \mathcal{S}_1$ and $\langle M, w, v' \rangle \in \mathcal{S}_2$. Then by $\langle M, v, w \rangle \models t_1$ and $\langle M, w, v' \rangle \models t_2$, $\langle M, v, v' \rangle \models t_1 \cdot t_2$, and hence $\langle \mathcal{S}_1 \cdot \mathcal{S}_2, \emptyset \rangle \models t_1 \cdot t_2$. (2)$(\Rightarrow)$: Let $\tilde{M} = \langle M, v, v' \rangle$. Note that, by $\tilde{M} \not\models t_1 \cdot t_2$, for every $w \in |M|$, $\langle M, v, w \rangle \not\models t_1$ or $\langle M, w, v' \rangle \not\models t_2$ holds. Then by letting $\mathcal{U}_1 = \{\langle M, v, w \rangle \mid w \in |M| \text{ and } \langle M, v, w \rangle \not\models t_1\}$ and $\mathcal{U}_2 = \{\langle M, w, v' \rangle \mid w \in |M| \text{ and } \langle M, w, v' \rangle \not\models t_2\}$, $\tilde{M} \in \mathcal{U}_1 \dagger \mathcal{U}_2$ (since $\tilde{M} \not\models t_1 \cdot t_2$), $\langle \emptyset, \mathcal{U}_1 \rangle \models t_1$ and $\langle \emptyset, \mathcal{U}_2 \rangle \models t_2$ hold. (2)$(\Leftarrow)$: By Proposition 4.1, it suffices to show $\langle \emptyset, \mathcal{U}_1 \dagger \mathcal{U}_2 \rangle \models t_1 \cdot t_2$. For every $\langle M, v, v' \rangle \in \mathcal{U}_1 \dagger \mathcal{U}_2$, $\langle M, v, w \rangle \in \mathcal{U}_1$ or $\langle M, w, v' \rangle \in \mathcal{U}_2$ holds for every $w \in |M|$. Then, by $\langle M, v, w \rangle \not\models t_1$ or $\langle M, w, v' \rangle \not\models t_2$ holds for every $w \in |M|$, $\langle M, v, v' \rangle \not\models t_1 \cdot t_2$, and hence $\langle \emptyset, \mathcal{U}_1 \dagger \mathcal{U}_2 \rangle \models t_1 \cdot t_2$. □

Theorem 4.3 *(sound- and completeness).* $\langle \mathcal{S}, \mathcal{U} \rangle \vdash t$ *iff* $\langle \mathcal{S}, \mathcal{U} \rangle \models t$.

Proof. $(\Rightarrow)$: By induction on the structure of the derivation tree of $\langle \mathcal{S}, \mathcal{U} \rangle \vdash t$. Case (WK): By I.H. and Proposition 4.1. Case (At): Trivial. Case $(\star)$ where $\star \in \{\bullet^{\pi}, \bullet^{-}, \cup, \cap, \cdot, \dagger\}$: By I.H. and Lemma 4.2$(\star)$. $(\Leftarrow)$: By induction on the structure of t. Case $t = \tilde{a}$: By applying (At). Case $t = t_1^{\pi}$: By Lemma 4.2$(\bullet^{\pi})$, there exists some $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle$ such that $\mathcal{S} \subseteq \mathcal{S}_1^{\pi}$, $\mathcal{U} \subseteq \mathcal{U}_1^{\pi}$, and $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \models t_1$. By I.H., $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \vdash t_1$. Thus by applying $(\bullet^{\pi})$ and (WK), $\langle \mathcal{S}, \mathcal{U} \rangle \vdash t$. Case $t = t_1^{-}$: By Lemma 4.2$(\bullet^{-})$, $\langle \mathcal{U}, \mathcal{S} \rangle \models t_1$. By I.H., $\langle \mathcal{U}, \mathcal{S} \rangle \vdash t_1$. Thus by applying $(\bullet^{-})$, $\langle \mathcal{S}, \mathcal{U} \rangle \vdash t$. Case $t = t_1 \cup t_2$: By Lemma 4.2$(\cup)$, there exist some $\mathcal{S}_1$ and $\mathcal{S}_2$ such that $\langle \mathcal{S}_1, \mathcal{U} \rangle \models t_1$ and $\langle \mathcal{S}_2, \mathcal{U} \rangle \models t_2$. By I.H., $\langle \mathcal{S}_1, \mathcal{U} \rangle \vdash t_1$ and $\langle \mathcal{S}_2, \mathcal{U} \rangle \vdash t_2$. Thus by applying $(\cup)$, $\langle \mathcal{S}, \mathcal{U} \rangle \vdash t$. Case $t = t_1 \cap t_2$: In the same manner as the case of $(\cup)$. Case $t = t_1 \cdot t_2$: By Lemma 4.2$(\cdot)$, there exist some $\mathcal{S}_1, \mathcal{S}_2, \mathcal{U}_1, \mathcal{U}_2$ such that $\mathcal{S} \subseteq \mathcal{S}_1 \cdot \mathcal{S}_2$, $\mathcal{U} \subseteq \mathcal{U}_1 \dagger \mathcal{U}_2$, $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \models t_1$ and $\langle \mathcal{S}_2, \mathcal{U}_2 \rangle \models t_2$. By I.H., $\langle \mathcal{S}_1, \mathcal{U}_1 \rangle \vdash t_1$ and $\langle \mathcal{S}_2, \mathcal{U}_2 \rangle \vdash t_2$. Thus by applying $(\cdot)$ and (WK), $\langle \mathcal{S}, \mathcal{U} \rangle \vdash t$. Case $t = t_1 \dagger t_2$: In the same manner as the case of $(\cdot)$. □

Example 4.4. Let M be the structure defined by $|M| := \{\mathsf{L}_1, \mathsf{L}_2, \mathsf{C}, \mathsf{R}_1, \mathsf{R}_2\}$ and $a^M := \{\langle \mathsf{L}_l, \mathsf{C} \rangle, \langle \mathsf{C}, \mathsf{R}_l \rangle \mid l \in [2]\}$. The following is a graphical description of M:



Then, $\langle \{\langle M, \mathsf{L}_i, \mathsf{R}_i \rangle \mid i \in [2]\}, \{\langle M, \mathsf{L}_1, \mathsf{L}_2 \rangle\} \rangle \vdash a \cdot a$ is shown as follows. As one can see from this example, we need the rule (WK).

$$\dfrac{\dfrac{}{\langle \{\langle M, \mathsf{L}_i, \mathsf{C} \rangle \mid i \in [2]\}, \{\langle M, \mathsf{L}_1, v \rangle \mid v \in |M| \setminus \{\mathsf{C}\}\} \rangle \vdash a}\ (\mathrm{At}) \quad \dfrac{}{\langle \{\langle M, \mathsf{C}, \mathsf{R}_j \rangle \mid j \in [2]\}, \{\langle M, \mathsf{C}, \mathsf{L}_2 \rangle\} \rangle \vdash a}\ (\mathrm{At})}{\dfrac{\langle \{\langle M, \mathsf{L}_i, \mathsf{R}_j \rangle \mid i, j \in [2]\}, \{\langle M, \mathsf{L}_1, \mathsf{L}_2 \rangle\} \rangle \vdash a \cdot a}{\langle \{\langle M, \mathsf{L}_i, \mathsf{R}_i \rangle \mid i \in [2]\}, \{\langle M, \mathsf{L}_1, \mathsf{L}_2 \rangle\} \rangle \vdash a \cdot a}\ (\mathrm{WK})}\ (\cdot)$$

4.1. Formula-size game and succinctness

In this subsection, we present a short description of how to use the formula-size game presented above.

Let $\mathrm{sz}_{\mathfrak{T}}(\mathcal{S},\mathcal{U}) := \min(\{\|t\| \mid t \in \mathfrak{T} \text{ and } \langle \mathcal{S},\mathcal{U}\rangle \models t\} \cup \{\infty\})$. If we can construct a function in the following, we can present a lower bound of $\mathrm{sz}_{\mathfrak{T}}(\mathcal{S},\mathcal{U})$ (Proposition 4.6) by using the formula-size game.

Definition 4.5 *(cf. [12]).* For a function $w\colon 2^{\mathbf{M}_2} \times 2^{\mathbf{M}_2} \to \mathbb{N} \cup \{\infty\}$, we define the following conditions (according to the derivation rule in Fig. 4).

(WK) $w(\mathcal{S},\mathcal{U}) \le w(\mathcal{S}\cup\mathcal{S}',\mathcal{U}\cup\mathcal{U}')$;
(At) $w(\mathcal{S},\mathcal{U}) \ge 1 \Longrightarrow \langle \mathcal{S},\mathcal{U}\rangle \not\models \tilde{a}$ (for any atomic term $\tilde{a}$);
($\bullet^{\pi}$) $w(\mathcal{S}_1^{\pi},\mathcal{U}_1^{\pi}) \le w(\mathcal{S}_1,\mathcal{U}_1)+1$;
($\bullet^{-}$) $w(\mathcal{S},\mathcal{U}) \le w(\mathcal{U},\mathcal{S})+1$;
($\cup$) $w(\mathcal{S}_1\cup\mathcal{S}_2,\mathcal{U}) \le w(\mathcal{S}_1,\mathcal{U})+w(\mathcal{S}_2,\mathcal{U})+1$;
($\cap$) $w(\mathcal{S},\mathcal{U}_1\cup\mathcal{U}_2) \le w(\mathcal{S},\mathcal{U}_1)+w(\mathcal{S},\mathcal{U}_2)+1$;
($\cdot$) $w(\mathcal{S}_1\cdot\mathcal{S}_2,\mathcal{U}_1\dagger\mathcal{U}_2) \le w(\mathcal{S}_1,\mathcal{U}_1)+w(\mathcal{S}_2,\mathcal{U}_2)+1$;
($\dagger$) $w(\mathcal{S}_1\dagger\mathcal{S}_2,\mathcal{U}_1\cdot\mathcal{U}_2) \le w(\mathcal{S}_1,\mathcal{U}_1)+w(\mathcal{S}_2,\mathcal{U}_2)+1$.

We say that w is *a weight function for* CoR if w satisfies all the above conditions. Also, we say that w is *a weight function for a term class* $\mathfrak{S}$ if w satisfies (WK) and $\mathfrak{S}$ is a subset of the term set generated by atomic terms satisfying the rule (At) w.r.t. w and operators $\star$ satisfying the rule ($\star$) w.r.t. w (e.g., we say that w is a weight function for PCoR if w satisfies (WK), (At) for any positive atomic term, and satisfies the conditions ($\bullet^{\pi}$), ($\cup$), ($\cap$), and ($\cdot$)).

Proposition 4.6. *Let w be a weight function for $\mathfrak{S}$. Then, for every term $t \in \mathfrak{S}$, $\langle \mathcal{S},\mathcal{U}\rangle \vdash t \Longrightarrow \|t\| \ge w(\mathcal{S},\mathcal{U})$. Thus, $\mathrm{sz}_{\mathfrak{S}}(\mathcal{S},\mathcal{U}) \ge w(\mathcal{S},\mathcal{U})$.*

Proof. By straightforward induction on the derivation tree of $\langle \mathcal{S},\mathcal{U}\rangle \vdash t$ using the conditions of "weight function for $\mathfrak{S}$" (in Definition 4.5). □

Also, the following proposition is useful to show succinctness results (for example, to prove Theorem 5.1, we will use this proposition with $\mathcal{F} = \mathcal{O}(n)$ and $\mathcal{G} = 2^{o(n)}$).

Proposition 4.7. *If there exists some sequence $\langle \mathcal{S}^{(n)},\mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ such that $\langle \mathrm{sz}_{\mathfrak{T}}(\mathcal{S}^{(n)},\mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \in \mathcal{F}$ and $\langle \mathrm{sz}_{\mathfrak{S}}(\mathcal{S}^{(n)},\mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \not\le_{\mathrm{b}} \mathcal{G}\circ\mathcal{F}$ (let us recall the $\le_{\mathrm{b}}$ in Definition 2.6), then $\mathfrak{T} \not\le^{\mathcal{G}}_{\mathbf{M}_2} \mathfrak{S}$.*

Proof. Assume, toward contradiction, $\mathfrak{T} \le^{\mathcal{G}}_{\mathbf{M}_2} \mathfrak{S}$. Let t_n be a term of size $\|t_n\| = \mathrm{sz}_{\mathfrak{T}}(\mathcal{S}^{(n)},\mathcal{U}^{(n)})$ such that $\langle \mathcal{S}^{(n)},\mathcal{U}^{(n)}\rangle \models t_n$. By the assumption, there exists some function $g \in \mathcal{G}$ such that for every t_n, there exists some $s_n \in \mathfrak{S}$ of size $\|s_n\| \le g(\|t_n\|)$ such that $s_n =_{\mathbf{M}_2} t_n$. Therefore, $\mathrm{sz}_{\mathfrak{S}}(\mathcal{S}^{(n)},\mathcal{U}^{(n)}) \le \|s_n\| \le g(\mathrm{sz}_{\mathfrak{T}}(\mathcal{S}^{(n)},\mathcal{U}^{(n)}))$, and thus $\langle \mathrm{sz}_{\mathfrak{S}}(\mathcal{S}^{(n)},\mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \le_{\mathrm{b}} \mathcal{G}\circ\mathcal{F}$, so it reaches a contradiction. Thus, $\mathfrak{T} \not\le^{\mathcal{G}}_{\mathbf{M}_2} \mathfrak{S}$. □

Combining it with Proposition 4.6, we will use this proposition in the following style.

Proposition 4.8. *If there exist some sequence $\langle w^{(n)}\rangle_{n\in\mathbb{N}_+}$ of weight functions for $\mathfrak{S}$ and some sequence $\langle \mathcal{S}^{(n)},\mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ such that $\langle \mathrm{sz}_{\mathfrak{T}}(\mathcal{S}^{(n)},\mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \in \mathcal{F}$ and $\langle w^{(n)}(\mathcal{S}^{(n)},\mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \not\le_{\mathrm{b}} \mathcal{G}\circ\mathcal{F}$, then $\mathfrak{T} \not\le^{\mathcal{G}}_{\mathbf{M}_2} \mathfrak{S}$.*

Proof. By Proposition 4.7 and 4.6. □

5. PCoR is exponentially less succinct than EP^3

In this section, we show the following theorem. Namely the exponential blowup of the translation from EP^3 to PCoR given in Section 3 is unavoidable. In this section, we assume that $\mathscr{A}$ is a countably infinite set.

Theorem 5.1. $\mathrm{EP}^3 \not\le^{2^{o(n)}}_{\mathbf{M}_2} \mathrm{PCoR}$.

We first present an example. For $n \in \mathbb{N}_+$, let $\mathfrak{t}_n$ be the following EP^3 term:

$$\mathfrak{t}_n := \left[\exists z.\ \bigwedge_{k\in[n]} \mathsf{a}_k(x,z) \vee \mathsf{a}_k(z,y)\right]_{x,y}.$$

Here, x, y, z are pairwise distinct and $\mathsf{a}_1, \mathsf{a}_2, \ldots$ are pairwise distinct binary relation symbols. Note that each $\mathfrak{t}_n$ is semantically equivalent to the following PCoR term:

$$\mathfrak{t}'_n = \bigcup_{\langle I,J\rangle \in \mathrm{Part}([n])} (\bigcap_{i\in I} \mathsf{a}_i) \cdot (\bigcap_{j\in J} \mathsf{a}_j).$$

Here, $\mathrm{Part}(X)$ denotes the set of all ordered partitions of size 2 of X with possibly empty sets (i.e., the set of all pairs $\langle I, J\rangle$ such that $I \cup J = X$ and $I \cap J = \emptyset$). However, unfortunately, this is not a $2^{o(n)}$-size translation, because $\|\mathfrak{t}_n\| \in \mathcal{O}(n)$ and $\#(\mathrm{Part}([n])) = 2^n$. In the following, we will show that there does not exist any $2^{o(n)}$-size translation from $\{\mathfrak{t}_n\}_{n\in\mathbb{N}_+}$ to equivalent PCoR terms.

5.1. Proof outline of Theorem 5.1

In this subsection, we present the proof outline of Theorem 5.1. We first remark the following proposition.

The set of *projection normal form* PCoR (written PNF-PCoR, for short) formulas is defined by the following grammar where $a \in \mathscr{A}$:

$$t, s \in \mathrm{Term}^{\text{PNF-PCoR}} ::= a^\smile \mid a \mid \mathbf{1} \mid \top \mid t \cap s \mid t \cdot s \mid \mathbf{0} \mid t \cup s.$$

Proposition 5.2. PCoR $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ *PNF*-PCoR.

Proof. First, we replace each sub-term $t^{\{1\mapsto 1,2\mapsto 2\}}$ with t; $t^{\{1\mapsto 1,2\mapsto 1\}}$ with $(t \cap \mathbf{1}) \cdot \top$; and $t^{\{1\mapsto 2,2\mapsto 2\}}$ with $\top \cdot (t \cap \mathbf{1})$, respectively. Then, the π of each sub-term t is converse. Secondly, we push converse operators deeper into the term by the following rewriting rules: $(t^\smile)^\smile \rightsquigarrow t$; $\mathbf{1}^\smile \rightsquigarrow \mathbf{1}$; $\top^\smile \rightsquigarrow \top$; $(t \cap s)^\smile \rightsquigarrow t^\smile \cap s^\smile$; $(t \cdot s)^\smile \rightsquigarrow s^\smile \cdot t^\smile$; $\mathbf{0}^\smile \rightsquigarrow \mathbf{0}$; $(t \cup s)^\smile \rightsquigarrow t^\smile \cup s^\smile$. □

To prove Theorem 5.1, it suffices to show the following.

Lemma 5.3. $\mathrm{EP}^3 \not\leq^{2^{o(n)}}_{\mathbf{M}_2}$ *PNF*-PCoR.

Proof of Lemma 5.3 $\Longrightarrow$ Theorem 5.1. We show the contraposition. By Proposition 2.7 and $\mathrm{EP}^3 \leq^{2^{o(n)}}_{\mathbf{M}_2}$ PCoR $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ PNF-PCoR, we have $\mathrm{EP}^3 \leq^{\mathcal{O}(n)\circ 2^{o(n)}}_{\mathbf{M}_2}$ PNF-PCoR. Hence by $\mathcal{O}(n) \circ 2^{o(n)} \leq_{\mathrm{b}} 2^{o(n)}$, the statement of the theorem has been proved. □

We say that a function $w\colon 2^{\mathbf{M}_2} \times 2^{\mathbf{M}_2} \to \mathbb{N} \cup \{\infty\}$ is *a weight function for PNF*-PCoR if w satisfies the conditions (WK), (At), (∪), (∩), and (·) in Definition 4.5. Then by using Proposition 4.8, it suffices to show the following to prove Lemma 5.3.

Lemma 5.4. *There exist a sequence $\langle w^{(n)}\rangle_{n\in\mathbb{N}_+}$ of weight functions for PNF*-PCoR *and a sequence $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ such that (1) $\langle \mathrm{sz}_{\mathrm{EP}^3}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \in \mathcal{O}(n)$ and (2) $\langle w^{(n)}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \not\leq_{\mathrm{b}} 2^{o(n)}$.*

In the following subsection, we define weight functions for Lemma 5.4.

5.2. Proof of Lemma 5.4: construction of weight functions

To construct weight functions, we define the following function.

Definition 5.5. Let $R \subseteq \mathbf{M}_2^2$. The function $\mathrm{w}^R\colon 2^{\mathbf{M}_2} \times 2^{\mathbf{M}_2} \to \mathbb{N} \cup \{\infty\}$ is defined as follows: (1) If there exists a pair $\langle \tilde{M}, \tilde{N}\rangle \in \mathcal{S} \times \mathcal{U}$ such that $\tilde{M} \subseteq \tilde{N}$, then $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) := \infty$ (in the following, we call such pair $\langle \tilde{M}, \tilde{N}\rangle$ *substructure pair*); (2) Otherwise, $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) := \#((\mathcal{S}^{\cong} \times \mathcal{U}^{\cong}) \cap R)$. Here, $\mathcal{C}^{\cong}$ denotes the isomorphic closure of $\mathcal{C}$ (i.e., the minimal set containing $\mathcal{C}$ such that, if $M \in \mathcal{C}^{\cong}$ and $M \cong N$, then $N \in \mathcal{C}^{\cong}$).

We often write $\mathrm{S}^R(\mathcal{S}, \mathcal{U})$ for the set $(\mathcal{S}^{\cong} \times \mathcal{U}^{\cong}) \cap R$. If $\mathrm{w}^R(\mathcal{S}, \mathcal{U})$ is finite, then $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) = \#(\mathrm{S}^R(\mathcal{S}, \mathcal{U}))$. The function w^R has the following properties for ∞.

Lemma 5.6.

(WK) $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) = \infty \Longrightarrow \mathrm{w}^R(\mathcal{S} \cup \mathcal{S}', \mathcal{U} \cup \mathcal{U}') = \infty$;
(At) $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) = \infty \Longrightarrow \langle \mathcal{S}, \mathcal{U}\rangle \not\models \tilde{a}$;
(∪) $\mathrm{w}^R(\mathcal{S}_1 \cup \mathcal{S}_2, \mathcal{U}) = \infty \Longleftrightarrow \mathrm{w}^R(\mathcal{S}_1, \mathcal{U}) = \infty$ *or* $\mathrm{w}^R(\mathcal{S}_2, \mathcal{U}) = \infty$;
(∩) $\mathrm{w}^R(\mathcal{S}, \mathcal{U}_1 \cup \mathcal{U}_2) = \infty \Longleftrightarrow \mathrm{w}^R(\mathcal{S}, \mathcal{U}_1) = \infty$ *or* $\mathrm{w}^R(\mathcal{S}, \mathcal{U}_2) = \infty$;
(·) $\mathrm{w}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) = \infty \Longrightarrow \mathrm{w}^R(\mathcal{S}_1, \mathcal{U}_1) = \infty$ *or* $\mathrm{w}^R(\mathcal{S}_2, \mathcal{U}_2) = \infty$.

Proof. (WK): Trivial. (At): Trivial. (∪): By that, $(\mathcal{S}_1 \cup \mathcal{S}_2) \times \mathcal{U}$ contains a substructure pair if and only if $\mathcal{S}_1 \times \mathcal{U}$ or $\mathcal{S}_2 \times \mathcal{U}$ contains a substructure pair. (∩): In the same manner as the case of (∪). (·): Let $\langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle$ be a substructure pair in $\langle \mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2\rangle$ and let v'' be a node such that $\langle M, v, v''\rangle \in \mathcal{S}_1$ and $\langle M, v'', v'\rangle \in \mathcal{S}_2$. We take w'' such that $\langle M, v, v', v''\rangle \subseteq \langle N, w, w', w''\rangle$. Then by $\langle N, w, w'\rangle \in \mathcal{U}_1 \dagger \mathcal{U}_2$, $\langle\langle M, v, v''\rangle, \langle N, w, w''\rangle\rangle \in \mathcal{S}_1 \times \mathcal{U}_1$ or $\langle\langle M, v'', v'\rangle, \langle N, w'', w'\rangle\rangle \in \mathcal{S}_2 \times \mathcal{U}_2$ holds. Therefore, $\mathrm{w}^R(\mathcal{S}_1, \mathcal{U}_1) = \infty$ or $\mathrm{w}^R(\mathcal{S}_2, \mathcal{U}_2) = \infty$. □

Also, this function has the following properties.

Lemma 5.7.

(WK) $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) \leq \mathrm{w}^R(\mathcal{S} \cup \mathcal{S}', \mathcal{U} \cup \mathcal{U}')$.
(At;⊤)(At;0) $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) \geq 1 \Longrightarrow \langle \mathcal{S}, \mathcal{U}\rangle \not\models \tilde{a}$ *where* $\tilde{a}$ *is* $\top$ *or* $\mathbf{0}$.
(∪) $\mathrm{w}^R(\mathcal{S}_1 \cup \mathcal{S}_2, \mathcal{U}) \leq \mathrm{w}^R(\mathcal{S}_1, \mathcal{U}) + \mathrm{w}^R(\mathcal{S}_2, \mathcal{U})$.
(∩) $\mathrm{w}^R(\mathcal{S}, \mathcal{U}_1 \cup \mathcal{U}_2) \leq \mathrm{w}^R(\mathcal{S}, \mathcal{U}_1) + \mathrm{w}^R(\mathcal{S}, \mathcal{U}_2)$.

Proof. By Lemma 5.6, the rest case is that the value on the left-hand side is not ∞. (WK): Trivial. (At): Trivial. (∪): By the following:

$$\begin{aligned}(\mathcal{S}^{\cong} \times \mathcal{U}^{\cong}) \cap R &= ((\mathcal{S}_1 \cup \mathcal{S}_2)^{\cong} \times \mathcal{U}^{\cong}) \cap R \\ &= ((\mathcal{S}_1^{\cong} \cup \mathcal{S}_2^{\cong}) \times \mathcal{U}^{\cong}) \cap R \\ &= ((\mathcal{S}_1^{\cong} \times \mathcal{U}^{\cong}) \cap R) \cup ((\mathcal{S}_2^{\cong} \times \mathcal{U}^{\cong}) \cap R).\end{aligned}$$

(∩): In the same manner as the case of (∪). □

For deriving other inequalities, we introduce the following notion.

Definition 5.8. We say that $R \subseteq \mathbf{M}_2^2$ is *good for PNF*-PCoR if $\#(\{\tilde{N} \mid \langle \tilde{M}, \tilde{N}\rangle \in R \text{ for some } \tilde{M}\}) \leq 1$ and the following hold.

(At;$a^{\smile}$)(At;a)(At;1) For every $\langle \tilde{M}, \tilde{N}\rangle \in R$, $\tilde{M} \models \tilde{a}$ iff $\tilde{N} \models \tilde{a}$ where $\tilde{a}$ is $a^{\smile}$, a, or $\mathbf{1}$.
($\bullet^{\{1\mapsto 1,2\mapsto 1\}}$) For every $\langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle \in R$, $\langle M, v, v\rangle \subseteq \langle N, w, w\rangle$.
($\bullet^{\{1\mapsto 2,2\mapsto 2\}}$) For every $\langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle \in R$, $\langle M, v', v'\rangle \subseteq \langle N, w', w'\rangle$.
(·) For every distinct pair $\langle\langle M_1, v_1, v_1'\rangle, \langle N, w, w'\rangle\rangle, \langle\langle M_2, v_2, v_2'\rangle, \langle N, w, w'\rangle\rangle \in R$ and every $v_1'' \in |M_1| \setminus \{v_1, v_1'\}$ and $v_2'' \in |M_2| \setminus \{v_2, v_2'\}$, there exists some $w'' \in |N|$ such that $\langle M_i, v_i, v_i''\rangle \subseteq \langle N, w, w''\rangle$ holds for some $i \in [2]$ and $\langle M_i, v_i'', v_i'\rangle \subseteq \langle N, w'', w'\rangle$ holds for some $i \in [2]$.

If R is good for PNF-PCoR, the following holds.

Lemma 5.9. *Assume that* R *is good for PNF*-PCoR.

(1) *Let* $\langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle \in \mathrm{S}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2)$. *If* $\mathrm{w}^R(\mathcal{S}_j, \mathcal{U}_j) \neq \infty$ *for each* $j \in [2]$, *then there exists some* $v'' \in |M| \setminus \{v, v'\}$ *such that* $\langle M, v, v''\rangle \in \mathcal{S}_1^{\cong}$ *and* $\langle M, v'', v'\rangle \in \mathcal{S}_2^{\cong}$.
(2) *For each* $i \in [2]$, *let* $r^{(i)} \in \mathrm{S}^R(\mathcal{S}_1^{(i)} \cdot \mathcal{S}_2^{(i)}, \mathcal{U}_1^{(i)} \dagger \mathcal{U}_2^{(i)})$. *If* $r^{(1)} \neq r^{(2)}$, *then* $\mathrm{w}^R(\bigcup_{i\in[2]} \mathcal{S}_j^{(i)}, \bigcup_{i\in[2]} \mathcal{U}_j^{(i)}) = \infty$ *for some* $j \in [2]$.
(3) *If* $\mathrm{w}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) \geq 2$, *then* $\mathrm{w}^R(\mathcal{S}_j, \mathcal{U}_j) = \infty$ *for some* $j \in [2]$.

Proof. (1): By $\langle M, v, v'\rangle \in (\mathcal{S}_1 \cdot \mathcal{S}_2)^{\cong}$, there exists some $v'' \in |M|$ such that $\langle M, v, v''\rangle \in \mathcal{S}_1^{\cong}$ and $\langle M, v'', v'\rangle \in \mathcal{S}_2^{\cong}$. By $\langle N, w, w'\rangle \in (\mathcal{U}_1 \dagger \mathcal{U}_2)^{\cong}$, for every $w'' \in |N|$, $\langle N, w, w''\rangle \in \times \mathcal{U}_1^{\cong}$ or $\langle N, w'', w'\rangle \in \mathcal{U}_2^{\cong}$. If we assume $v'' = v$, then by letting $w'' = w$ in the above, (a) $\langle\langle M, v, v\rangle, \langle N, w, w\rangle\rangle \in \mathcal{S}_1^{\cong} \times \mathcal{U}_1^{\cong}$ or (b) $\langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle \in \mathcal{S}_2^{\cong} \times \mathcal{U}_2^{\cong}$ holds. In both cases, it reaches a contradiction. If we assume (a), then by $\langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle \in R$ and Definition 5.8($\bullet^{\{1\mapsto 1,2\mapsto 1\}}$), we have $\langle M, v, v\rangle \subseteq \langle N, w, w\rangle$; hence it contradicts to $\mathrm{w}^R(\mathcal{S}_1, \mathcal{U}_1) \neq \infty$. If we assume (b), then $r = \langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle \in \mathrm{S}^R(\mathcal{S}_2, \mathcal{U}_2)$, and thus it reaches a contradiction. Hence, $v'' \neq v$. In the same manner, if we assume that $v'' = v'$, by letting $w'' = w'$ in the above, it reaches a contradiction. Hence, $v'' \neq v'$. Therefore, $v'' \in |M| \setminus \{v, v'\}$.

(2): If $\mathrm{w}^R(\mathcal{S}_j^{(i)}, \mathcal{U}_j^{(i)}) = \infty$ for some $i \in [2]$ and $j \in [2]$, then by Lemma 5.6(WK), this case has been proved. Otherwise, $\mathrm{w}^R(\mathcal{S}_j^{(i)}, \mathcal{U}_j^{(i)}) \neq \infty$ for each $i \in [2]$ and $j \in [2]$. For each $i \in [2]$, let $r^{(i)} = \langle\langle M^{(i)}, v^{(i)}, (v')^{(i)}\rangle, \langle N, w, w'\rangle\rangle$ (note that $\#(\{\tilde{N} \mid \langle \tilde{M}, \tilde{N}\rangle \in R \text{ for some } \tilde{M}\}) \leq 1$). Then by (1), there exists some $(v'')^{(i)} \in |M^{(i)}| \setminus \{v^{(i)}, (v')^{(i)}\}$ such that $\langle M^{(i)}, v^{(i)}, (v'')^{(i)}\rangle \in (\mathcal{S}_1^{(i)})^{\cong}$ and $\langle M^{(i)}, (v'')^{(i)}, (v')^{(i)}\rangle \in (\mathcal{S}_2^{(i)})^{\cong}$. By $\langle N, w, w'\rangle \in (\mathcal{U}_1^{(i)} \dagger \mathcal{U}_2^{(i)})^{\cong}$, for every $w'' \in |N|$, $\langle N, w, w''\rangle \in (\mathcal{U}_1^{(i)})^{\cong}$ or $\langle N, w'', w'\rangle \in (\mathcal{U}_2^{(i)})^{\cong}$. Hence,

- $\langle\langle M^{(i)}, v^{(i)}, (v'')^{(i)}\rangle, \langle N, w, w''\rangle\rangle \in (\bigcup_{i\in[2]} \mathcal{S}_1^{(i)})^{\cong} \times (\bigcup_{i\in[2]} \mathcal{U}_1^{(i)})^{\cong}$ for each $i \in [2]$; or
- $\langle\langle M^{(i)}, (v'')^{(i)}, (v')^{(i)}\rangle, \langle N, w'', w'\rangle\rangle \in (\bigcup_{i\in[2]} \mathcal{S}_2^{(i)})^{\cong} \times (\bigcup_{i\in[2]} \mathcal{U}_2^{(i)})^{\cong}$ for each $i \in [2]$.

Moreover, by Definition 5.8(·), there exists some $w'' \in |N|$ such that

- $\langle M^{(i)}, v^{(i)}, (v'')^{(i)}\rangle \subseteq \langle N, w, w''\rangle$ holds for some $i \in [2]$; and
- $\langle M^{(i)}, (v'')^{(i)}, (v')^{(i)}\rangle \subseteq \langle N, w'', w'\rangle$ holds for some $i \in [2]$.

Combining them, $(\bigcup_{i\in[2]} \mathcal{S}_1^{(i)})^{\cong} \times (\bigcup_{i\in[2]} \mathcal{U}_1^{(i)})^{\cong}$ or $(\bigcup_{i\in[2]} \mathcal{S}_2^{(i)})^{\cong} \times (\bigcup_{i\in[2]} \mathcal{U}_2^{(i)})^{\cong}$ should contain some substructure pair, and thus $\mathrm{w}^R(\bigcup_{i\in[2]} \mathcal{S}_1^{(i)}, \bigcup_{i\in[2]} \mathcal{U}_1^{(i)}) = \infty$ or $\mathrm{w}^R(\bigcup_{i\in[2]} \mathcal{S}_2^{(i)}, \bigcup_{i\in[2]} \mathcal{U}_2^{(i)}) = \infty$.

(3): By (2) with $\mathcal{S}_j^{(1)} = \mathcal{S}_j^{(2)}$ and $\mathcal{U}_j^{(1)} = \mathcal{U}_j^{(2)}$ for each $j \in [2]$. □

Using this lemma, we can obtain the remaining conditions of weight function for PNF-PCoR as follows.

Lemma 5.10. *If R is good for PNF-*PCoR*, then the following hold.*

(At) $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) \geq 1 \Longrightarrow \langle \mathcal{S}, \mathcal{U}\rangle \not\models \tilde{a}$.
(·) $\mathrm{w}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) \leq \mathrm{w}^R(\mathcal{S}_1, \mathcal{U}_1) + \mathrm{w}^R(\mathcal{S}_2, \mathcal{U}_2) + 1$.

Proof. By Lemma 5.6, the rest case is that the value on the left-hand side is not ∞. (At): Then, there exists some $\langle \tilde{M}, \tilde{N}\rangle \in \mathrm{S}^R(\mathcal{S}, \mathcal{U})$. By Lemma 5.7(At) and Definition 5.8(At), $\langle \tilde{M}, \tilde{N}\rangle \not\models \tilde{a}$. Thus, $\langle \mathcal{S}, \mathcal{U}\rangle \not\models \tilde{a}$. (·): By $\mathrm{w}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) \neq \infty$ and Lemma 5.6, $\mathrm{w}^R(\mathcal{S}_j, \mathcal{U}_j) \neq \infty$ for each $j \in [2]$. We assume, toward contradiction, that $\mathrm{w}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) \geq \mathrm{w}^R(\mathcal{S}_1, \mathcal{U}_1) + \mathrm{w}^R(\mathcal{S}_2, \mathcal{U}_2) + 2$. Clearly, $\mathrm{w}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) \geq 2$. Thus by Lemma 5.9 (3), $\mathrm{w}^R(\mathcal{S}_j, \mathcal{U}_j) = \infty$ for some $j \in [2]$, but it reaches a contradiction. □

Lemma 5.11. *If R is good for PNF-*PCoR*, then w^R is a weight function for PNF-*PCoR.

Proof. By Lemma 5.7 and 5.10. □

Using this lemma, to prove Lemma 5.4, it suffices to show the following.

Lemma 5.12. *There exists a sequence $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ such that*

(1) *$\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ is good for PNF-*PCoR*, and*
(2) *$\langle \mathrm{sz}_{\mathrm{EP}^3}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \in \mathcal{O}(n)$,*
(3) *$\langle \#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \not\leq_{\mathrm{b}} 2^{o(n)}$.*

Proof of Lemma 5.12 $\Longrightarrow$ Lemma 5.4. Let $w^{(n)} := \mathrm{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}$ for $n \in \mathbb{N}_+$. Then, $w^{(n)}$ is a weight function for PNF-PCoR by (1) and Lemma 5.11. (1): By (2). (2): By $\mathrm{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)}) = \#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)})$ and (3). □

In the following subsection, we construct a sequence $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ such that $\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ is good for PNF-PCoR for Lemma 5.12.

5.3. Proof of Lemma 5.12: construction of structures

We define the sequence $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ of structures as follows.

Definition 5.13. We let $\mathcal{S}^{(n)} := \{\tilde{\mathrm{S}}_I^{(n)} \mid \emptyset \subsetneq I \subsetneq [n]\}$ and $\mathcal{U}^{(n)} := \{\tilde{\mathrm{U}}^{(n)}\}$. Here, $\tilde{\mathrm{S}}_I^{(n)}$ and $\tilde{\mathrm{U}}^{(n)}$ are defined as follows.

($\tilde{\mathrm{S}}_I^{(n)}$) Let $n \in \mathbb{N}$ and $I \subseteq [n]$. Then, let $\tilde{\mathrm{S}}_I^{(n)}$ be the 2-pointed structure $\langle \mathrm{S}_I^{(n)}, \mathsf{L}, \mathsf{R}\rangle$ defined by: $|\mathrm{S}_I^{(n)}| := \{\mathsf{L}, \mathsf{C}, \mathsf{R}\}$ and $\mathsf{a}_i^{\mathrm{S}_I^{(n)}} := \{\langle \mathsf{L}, \mathsf{C}\rangle \mid i \in I\} \cup \{\langle \mathsf{C}, \mathsf{R}\rangle \mid i \in [n] \setminus I\}$.
($\tilde{\mathrm{U}}^{(n)}$) Let $n \in \mathbb{N}$. Then, let $\tilde{\mathrm{U}}^{(n)}$ be the 2-pointed structure $\langle \mathrm{U}^{(n)}, \mathsf{L}, \mathsf{R}\rangle$ defined by: $|\mathrm{U}^{(n)}| := \{\mathsf{L}, \mathsf{R}\} \cup \{\mathsf{l}_{\langle I,J\rangle}, \mathsf{c}_{\langle I,J\rangle}, \mathsf{r}_{\langle I,J\rangle} \mid \emptyset \subsetneq I, J, I \cup ([n] \setminus J) \subsetneq [n]\}$ and $\mathsf{a}_i^{\mathrm{U}^{(n)}} := (\{\langle \mathsf{L}, \mathsf{c}_{\langle I,J\rangle}\rangle \mid i \in I\} \cup \{\langle \mathsf{c}_{\langle I,J\rangle}, \mathsf{r}_{\langle I,J\rangle}\rangle \mid i \in [n] \setminus I\} \cup \{\langle \mathsf{l}_{\langle I,J\rangle}, \mathsf{c}_{\langle I,J\rangle}\rangle \mid i \in J\} \cup \{\langle \mathsf{c}_{\langle I,J\rangle}, \mathsf{R}\rangle \mid i \in [n] \setminus J\}) \cap |\mathrm{U}^{(n)}|^2$.

Fig. 5 presents a graphical description of $\tilde{\mathrm{U}}^{(n)}$ where $n = 2$.

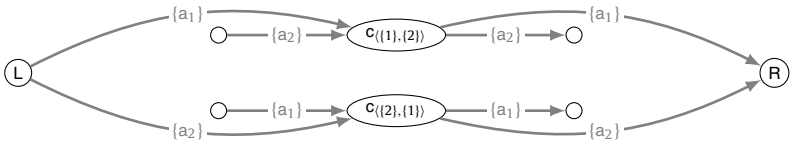


Fig. 5. Graphical description of $\tilde{\mathfrak{U}}_n$ where $n = 2$.

Proposition 5.14. *$\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ is good for PNF-*PCoR.

Proof. (At): It is because $\tilde{\mathsf{S}}_I^{(n)} \not\models \tilde{a}$ and $\tilde{\mathfrak{U}}^{(n)} \not\models \tilde{a}$, where $\tilde{a}$ is a, $a^{\smile}$, or $\mathbf{1}$. ($\bullet^{\{1\mapsto 1,2\mapsto 1\}}$) and ($\bullet^{\{1\mapsto 2,2\mapsto 2\}}$): Immediate from the form of the structures. ($\cdot$): Let $\langle\langle \mathsf{S}_I^{(n)}, \mathsf{L}, \mathsf{R}\rangle, \langle \mathfrak{U}^{(n)}, \mathsf{L}, \mathsf{R}\rangle\rangle, \langle\langle \mathsf{S}_J^{(n)}, \mathsf{L}, \mathsf{R}\rangle, \langle \mathfrak{U}^{(n)}, \mathsf{L}, \mathsf{R}\rangle\rangle \in \mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ be distinct and let w be $\mathsf{c}_{\langle I,J\rangle}$ or $\mathsf{c}_{\langle J,I\rangle}$ (note that one of the two is in $|\mathfrak{U}^{(n)}|$, because $I \cup ([n] \setminus J) \subsetneq [n]$ or $J \cup ([n] \setminus I) \subsetneq [n]$). If $w = \mathsf{c}_{\langle I,J\rangle}$, by $\langle \mathsf{S}_I^{(n)}, \mathsf{L}, \mathsf{C}\rangle \subseteq \langle \mathfrak{U}^{(n)}, \mathsf{L}, \mathsf{c}_{\langle I,J\rangle}\rangle$ and $\langle \mathsf{S}_J^{(n)}, \mathsf{C}, \mathsf{R}\rangle \subseteq \langle \mathfrak{U}^{(n)}, \mathsf{c}_{\langle I,J\rangle}, \mathsf{R}\rangle$. If $w = \mathsf{c}_{\langle J,I\rangle}$, by $\langle \mathsf{S}_J^{(n)}, \mathsf{L}, \mathsf{C}\rangle \subseteq \langle \mathfrak{U}^{(n)}, \mathsf{L}, \mathsf{c}_{\langle J,I\rangle}\rangle$ and $\langle \mathsf{S}_I^{(n)}, \mathsf{C}, \mathsf{R}\rangle \subseteq \langle \mathfrak{U}^{(n)}, \mathsf{c}_{\langle J,I\rangle}, \mathsf{R}\rangle$. □

Proof of Lemma 5.12. (1): By Proposition 5.14. (2): By $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle \models \mathfrak{t}_n$ and $\langle \|\mathfrak{t}_n\|\rangle_{n\in\mathbb{N}_+} \in \mathcal{O}(n)$. (3): By $\#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}) = 2^n - 2$. □

6. Extensions of Theorem 5.1

In this section, we present some extensions of Theorem 5.1.

6.1. Σ_1^{CoR} *is exponentially less succinct than* EP^3

In this subsection, we show the following. Namely, Theorem 5.1 can be extended from PCoR to Σ_1^{CoR} (see Section 3.1 for Σ_1^{CoR}).

Theorem 6.1. $\mathrm{EP}^3 \not\leq^{2^{o(n)}}_{\mathbf{M}_2} \Sigma_1^{\mathrm{CoR}}$.

We define the set of *PNF-NNF-*Σ_1^{CoR} (as with PNF-PCoR) as follows:

$$t, s \in \mathrm{Term}^{\text{PNF-NNF-PCoR}} ::= a^{\smile} \mid a \mid (a^{-})^{\smile} \mid a^{-} \mid \mathbf{1} \mid \top \mid t \cap s \mid t \cdot s \mid \mathbf{0} \mid t \cup s.$$

Then the following holds.

Proposition 6.2. $\Sigma_1^{\mathrm{CoR}} \leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ *PNF-NNF-*Σ_1^{CoR}.

Proof sketch. By taking the NNF for CoR terms as with Lemma 3.3 for FO^k and taking the PNF as with Proposition 5.2. □

From this, it suffices to show the following (as with Lemma 5.3).

Lemma 6.3. $\mathrm{EP}^3 \not\leq^{2^{o(n)}}_{\mathbf{M}_2}$ *PNF-NNF-*Σ_1^{CoR}.

Lemma 6.4. *If R is good for PNF-*PCoR*, then* w^R *is a weight function for PNF-NNF-*Σ_1^{CoR}.

Proof. By Definition 5.8, even for $\tilde{a} = (a^{-})^{\smile}$ and $\tilde{a} = a^{-}$, we have the condition (At) of Definition 5.8: For every $\langle \tilde{M}, \tilde{N}\rangle \in R$, $\tilde{M} \models \tilde{a}$ iff $\tilde{N} \models \tilde{a}$. Hence, even for $\tilde{a} = (a^{-})^{\smile}$ and $\tilde{a} = a^{-}$, we have the condition (At) of weight function (Definition 4.5): $\mathrm{w}^R(\mathcal{S}, \mathcal{U}) \geq 1 \Longrightarrow \langle \mathcal{S}, \mathcal{U}\rangle \not\models \tilde{a}$. □

Proof of Lemma 6.3. By using the same sequence $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ (Definition 5.13) and the same weight function $\mathrm{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}$. Note that by Lemma 6.4, $\mathrm{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}$ is not only a weight function for PNF-PCoR, but also a weight function for PNF-NNF-Σ_1^{CoR}. □

6.2. Over structures with one binary relation symbol

In this subsection, we consider the case over structures with one binary relation symbol. Even in this case, there does not exist any polynomial-size translation. More precisely, the following holds.

Theorem 6.5. *If* $\#(\mathscr{A}) = 1$, *then* $\mathrm{EP}^3 \not\leq^{2^{o(\sqrt{n})}}_{\mathbf{M}_2} \Sigma_1^{\mathrm{CoR}}$.

By Lemma 6.3, it suffices to show the following (note that $\mathcal{O}(n) \circ 2^{o(\sqrt{n})} \leq_{\mathrm{b}} 2^{o(\sqrt{n})}$).

Lemma 6.6. *If* $\#(\mathscr{A}) = 1$, *then* $\mathrm{EP}^3 \not\leq^{2^{o(\sqrt{n})}}_{\mathbf{M}_2}$ *PNF-NNF-*Σ_1^{CoR}.

To prove it, we modify the EP^3 term $\mathfrak{t}_n$ as follows:

$$\mathfrak{t}_n := \left[\exists z. \bigwedge_{k \in [n]} \mathsf{a}^k(x, z) \vee \mathsf{a}^k(z, y) \right]_{x,y}.$$

Here, $a^k(x, y) := \exists z. a(x, z) \wedge a^{k-1}(z, y)$ if $k > 1$; and $a^k(x, y) := a(x, y)$ if $k = 1$, where z is a variable distinct from x and y.

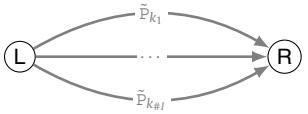
Also, we modify $\mathcal{S}^{(n)}$ and $\mathcal{U}^{(n)}$ as follows.

Definition 6.7. We let $\mathcal{S}^{(n)} := \{\tilde{\mathtt{S}}_I^{(n)} \mid \emptyset \subsetneq I \subsetneq [n]\}$ and $\mathcal{U}^{(n)} := \{\tilde{\mathtt{U}}^{(n)}\}$. Here, the 2-pointed structures $\tilde{\mathtt{S}}_I^{(n)}$ and $\tilde{\mathtt{U}}^{(n)}$ are defined as follows.

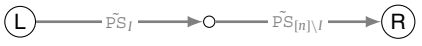
($\tilde{\mathtt{P}}_k$) Let $k \in \mathbb{N}$. Then, let $\tilde{\mathtt{P}}_k$ be the 2-pointed structure $\langle \mathtt{P}_k, \mathsf{L}, \mathsf{R} \rangle$ defined by $|\mathtt{P}_k| := \{\mathsf{C}_0, \ldots \mathsf{C}_k\}$, $\mathsf{a}^{\mathtt{P}_k} := \{\langle \mathsf{C}_{i-1}, \mathsf{C}_i \rangle \mid i \in [k]\}$, and $\langle \mathsf{L}, \mathsf{R} \rangle := \langle \mathsf{C}_0, \mathsf{C}_k \rangle$. The following is a graphical description of $\tilde{\mathtt{P}}_k$.



($\tilde{\mathtt{PS}}_I$) Let $I = \{k_1, \ldots, k_{\#I}\} \subseteq \mathbb{N}$ be a finite set. Then, let $\tilde{\mathtt{PS}}_I$ be the 2-pointed structure $\langle \mathtt{PS}_I, \mathsf{L}, \mathsf{R} \rangle$ defined by $\mathtt{PS}_I := (\biguplus_{i \in [\#I]} \mathtt{P}_{k_i})/\sim$ and $\langle \mathsf{L}, \mathsf{R} \rangle := \langle [\langle i, \mathsf{L} \rangle]_\sim, [\langle i, \mathsf{R} \rangle]_\sim \rangle$ (where i is any). Here, $\sim$ is the minimum equivalence relation satisfying $\langle i, \mathsf{L} \rangle \sim \langle j, \mathsf{L} \rangle$ and $\langle i, \mathsf{R} \rangle \sim \langle j, \mathsf{R} \rangle$. Also, let us recall that $\uplus$ means the disjoint union of structures (see the preliminaries). The following is a graphical description of $\tilde{\mathtt{PS}}_I$.



($\tilde{\mathtt{S}}_I^{(n)}$) Let $n \in \mathbb{N}$ and let $I \subseteq [n]$ be a finite set. Then, let $\tilde{\mathtt{S}}_I^{(n)}$ be the 2-pointed structure $\langle \mathtt{S}_I^{(n)}, \mathsf{L}, \mathsf{R} \rangle$ defined by $\mathtt{S}_I^{(n)} := (\mathtt{PS}_I \uplus \mathtt{PS}_{[n] \setminus I})/\sim$ and $\langle \mathsf{L}, \mathsf{R} \rangle := \langle [\langle 1, \mathsf{L} \rangle]_\sim, [\langle 2, \mathsf{R} \rangle]_\sim \rangle$. Here, $\sim$ is the minimum equivalence relation satisfying $\langle 1, \mathsf{R} \rangle \sim \langle 2, \mathsf{L} \rangle$. The following is a graphical description of $\tilde{\mathtt{S}}_I^{(n)}$.



($(\tilde{\mathtt{SP}}_{I,J}^{v,w})^{(n)}$) Let $n \in \mathbb{N}$, let $I, J \subseteq [n]$ be finite sets, and let $v \in |\mathtt{S}_I^{(n)}|$ and $w \in |\mathtt{S}_J^{(n)}|$ be vertices. Then, let $(\tilde{\mathtt{SP}}_{I,J}^{v,w})^{(n)}$ be the 3-pointed structure $\langle (\mathtt{SP}_{I,J}^{v,w})^{(n)}, \mathsf{L}, \mathsf{R}, \mathsf{c} \rangle$ defined by $(\mathtt{SP}_{I,J}^{v,w})^{(n)} := (\mathtt{S}_I^{(n)} \uplus \mathtt{S}_J^{(n)})/\sim$ and $\langle \mathsf{L}, \mathsf{R}, \mathsf{c} \rangle := \langle [\langle 1, \mathsf{L} \rangle]_\sim, [\langle 2, \mathsf{R} \rangle]_\sim, [\langle 1, v \rangle]_\sim \rangle$. Here, $\sim$ is the minimum equivalence relation satisfying $\langle 1, v \rangle \sim \langle 2, w \rangle$. The following is a graphical description of $(\tilde{\mathtt{SP}}_{I,J}^{v,w})^{(n)}$.



($\tilde{\mathtt{U}}^{(n)}$) Let $n \in \mathbb{N}$ and let K be the set of all the pairwise distinct elements $\langle I_k, J_k, v_k, w_k \rangle$ $(k \in [\#K])$ such that $\emptyset \subsetneq I_k, J_k, I_k \cup ([n] \setminus J_k) \subsetneq [n]$, $v_k \in |\mathtt{S}_{I_k}^{(n)}| \setminus \{\mathsf{L}, \mathsf{R}\}$, and $w_k \in |\mathtt{S}_{J_k}^{(n)}| \setminus \{\mathsf{L}, \mathsf{R}\}$. Then, let $\tilde{\mathtt{U}}^{(n)}$ be the 2-pointed structure

$\langle \mathsf{U}^{(n)}, \mathsf{L}, \mathsf{R}\rangle$ defined by $\mathsf{U}^{(n)} := (\biguplus_{k\in[\#K]}(\mathsf{SP}^{v_k,w_k}_{I_k,J_k})^{(n)})/\sim$, $\langle \mathsf{L}, \mathsf{R}\rangle := \langle [\langle i, \mathsf{L}\rangle]_\sim, [\langle i, \mathsf{R}\rangle]_\sim\rangle$ (where i is any). Here, $\sim$ is the minimum equivalence relation satisfying $\langle i, \mathsf{L}\rangle \sim \langle j, \mathsf{L}\rangle$ and $\langle i, \mathsf{R}\rangle \sim \langle j, \mathsf{R}\rangle$.

Then, the following holds.

Lemma 6.8.

(1) $\langle \mathrm{sz}_{\mathrm{EP}^3}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \in \mathcal{O}(n^2)$;
(2) $\langle \#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \not\leq_{\mathrm{b}} 2^{o(n)}$; *and*
(3) $\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ *is good for PNF-*PCoR.

Proof. (1): By $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle \models \mathfrak{t}_n$ and $\langle \|\mathfrak{t}_n\|\rangle_{n\in\mathbb{N}_+} \in \mathcal{O}(n^2)$. (2): By $\#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}) = 2^n - 2$. (3): We show each condition. (At): Since $\tilde{\mathsf{S}}^{(n)}_I \not\models \tilde{a}$ always, where $\tilde{a}$ is a, $a^\smile$, or $\mathbf{1}$. ($\bullet^{\{1\mapsto 1,2\mapsto 1\}}$) and ($\bullet^{\{1\mapsto 2,2\mapsto 2\}}$): Immediate from the form of the structures. $(\cdot)$: Let $\langle\langle \mathsf{S}^{(n)}_I, \mathsf{L}, \mathsf{R}\rangle, \langle \mathsf{U}^{(n)}, \mathsf{L}, \mathsf{R}\rangle\rangle, \langle\langle \mathsf{S}^{(n)}_J, \mathsf{L}, \mathsf{R}\rangle, \langle \mathsf{U}^{(n)}, \mathsf{L}, \mathsf{R}\rangle\rangle \in \mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ be distinct. Let $v \in |\mathsf{S}^{(n)}_I| \setminus \{\mathsf{L}, \mathsf{R}\}$ and $w \in |\mathsf{S}^{(n)}_J| \setminus \{\mathsf{L}, \mathsf{R}\}$ be any. Then, let u be $\mathsf{c}^{v,w}_{I,J}$ or $\mathsf{c}^{w,v}_{J,I}$ (note that one of the two is in $|\mathsf{U}^{(n)}|$, because $I \cup ([n] \setminus J) \subsetneq [n]$ or $J \cup ([n] \setminus I) \subsetneq [n]$ always holds). If $w = \mathsf{c}^{v,w}_{I,J}$, by $\langle \mathsf{S}^{(n)}_I, \mathsf{L}, v\rangle \subseteq \langle \mathsf{U}^{(n)}, \mathsf{L}, u\rangle$ and $\langle \mathsf{S}^{(n)}_J, w, \mathsf{R}\rangle \subseteq \langle \mathsf{U}^{(n)}, u, \mathsf{R}\rangle$. If $w = \mathsf{c}^{w,v}_{J,I}$, by $\langle \mathsf{S}^{(n)}_J, \mathsf{L}, w\rangle \subseteq \langle \mathsf{U}^{(n)}, \mathsf{L}, u\rangle$ and $\langle \mathsf{S}^{(n)}_I, v, \mathsf{R}\rangle \subseteq \langle \mathsf{U}^{(n)}, u, \mathsf{R}\rangle$. □

Proof of Lemma 6.6. Since $\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ is good for PNF-PCoR (Lemma 6.8(3)), $\mathsf{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}$ is a weight function for PNF-NNF-Σ_1^{CoR} (Lemma 6.4). Let us assume that $\langle \mathrm{sz}_{\text{PNF-NNF-}\Sigma_1^{\mathrm{CoR}}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \leq_{\mathrm{b}} 2^{o(n)}$. Then, it contradicts Lemma 6.8(2) as follows:

$$\begin{aligned}\langle \#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} &= \langle \mathsf{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \\ &\leq_{\mathrm{b}} \langle \mathrm{sz}_{\text{PNF-NNF-}\Sigma_1^{\mathrm{CoR}}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} && (\mathsf{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}} \text{ is a weight function}) \\ &\leq_{\mathrm{b}} 2^{o(n)}.\end{aligned}$$

Therefore, $\langle \mathrm{sz}_{\text{PNF-PCoR}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \not\leq_{\mathrm{b}} 2^{o(n)}$. Hence, the statement of the theorem has been proved by Proposition 4.7 with $\mathcal{F} = \mathcal{O}(n^2)$ and $\mathcal{G} = 2^{o(\sqrt{n})}$ (note that $2^{o(\sqrt{n})} \circ \mathcal{O}(n^2) = 2^{o(\sqrt{n})\circ\mathcal{O}(n^2)} \leq_{\mathrm{b}} 2^{o(n)}$). □

6.3. On the succinctness for DAG-size

In this subsection, we show that we can strength Theorem 5.1 from (tree-)size to DAG-size (Theorem 6.12). Here, DAG abbreviates directed acyclic graph.

Definition 6.9. The *DAG-size* of a term t, written $\|t\|_{(\mathrm{DAG})}$, is defined as the number of subterms of t.

Example 6.10. For example, we have $\|(a\cdot a)\cup(a\cdot a)\|_{(\mathrm{DAG})} = 3$ by that the term has itself, $a\cdot a$, and a as a subterm. Meanwhile, $\|(a\cdot a)\cup(a\cdot a)\| = 7$. More generally, let t_n be the PCoR-term defined by $t_0 := a$ and $t_{n+1} := t_n \cdot t_n$. Then, we have $\|t_n\| \geq 2^n$, whereas $\|t_n\|_{(\mathrm{DAG})} = n + 1$.

As this example demonstrates, the following holds between the two notions of *size*, $\|\bullet\|$ and $\|\bullet\|_{(\mathrm{DAG})}$.

Proposition 6.11. *The following hold.*

(1) *For every* PCoR*-term* t, $\|t\|_{(\mathrm{DAG})} \leq \|t\| \leq 2^{\|t\|_{(\mathrm{DAG})}} - 1$.
(2) *For every* $f \in 2^{o(n)}$*, there is some term* t *such that* $\|t\| \not\leq f(\|t\|_{(\mathrm{DAG})})$.

Proof. (1): The both inequations are shown by induction on the structure of t. (2): By using the sequence $\langle t_n\rangle_{n\in\mathbb{N}_+}$ in Example 6.10. □

For a term class $\mathfrak{T}$, we explicitly write $\mathfrak{T}_{(\mathrm{DAG})}$ for the term class $\mathfrak{T}$ where the DAG-size $\|\bullet\|_{(\mathrm{DAG})}$ is used instead of $\|\bullet\|$. The goal of this subsection is to show the following.

Theorem 6.12 *(cf. Theorem 6.1).* $\mathrm{EP}^3 \not\leq^{2^{o(n)}}_{\mathbf{M}_2} \Sigma_1^{\mathrm{CoR}}{}_{(\mathrm{DAG})}$.

As with Lemma 6.3, it suffices to show the following lemma.

Lemma 6.13 *(cf. Lemma 6.3).* $\mathrm{EP}^3 \not\leq^{2^{o(n)}}_{\mathbf{M}_2}$ *PNF-NNF-*$\Sigma_1^{\mathrm{CoR}}{}_{(\mathrm{DAG})}$.

Let us recall w^R and S^R (Definition 5.5). To prove this lemma, we should consider the set $\mathrm{S}^R(\mathcal{S},\mathcal{U}) = (\mathcal{S}^{\cong} \times \mathcal{U}^{\cong}) \cap R$ instead of the number $\mathrm{w}^R(\mathcal{S},\mathcal{U}) = \#((\mathcal{S}^{\cong} \times \mathcal{U}^{\cong}) \cap R)$. We have the following inequations for $\mathrm{S}^R(\mathcal{S},\mathcal{U})$ (like Lemma 5.7 and 5.10 for $\mathrm{w}^R(\mathcal{S},\mathcal{U})$).

Lemma 6.14 *(cf. Lemma 5.7 and 5.10). Assume that R is good for PNF-*PCoR.

(WK) $\mathrm{S}^R(\mathcal{S},\mathcal{U}) \subseteq \mathrm{S}^R(\mathcal{S} \cup \mathcal{S}', \mathcal{U} \cup \mathcal{U}')$.
(At) $\mathrm{S}^R(\mathcal{S},\mathcal{U}) \neq \emptyset \Longrightarrow \langle \mathcal{S},\mathcal{U}\rangle \not\models \tilde{a}$ *where* $\tilde{a}$ *is* $\top$, $\mathbf{0}$, $a^{\smile}$, a, $(a^-)^{\smile}$, a^-, *or* $\mathbf{1}$.
(∪) $\mathrm{S}^R(\mathcal{S}_1 \cup \mathcal{S}_2, \mathcal{U}) \subseteq \mathrm{S}^R(\mathcal{S}_1,\mathcal{U}) \cup \mathrm{S}^R(\mathcal{S}_2,\mathcal{U})$.
(∩) $\mathrm{S}^R(\mathcal{S}, \mathcal{U}_1 \cup \mathcal{U}_2) \subseteq \mathrm{S}^R(\mathcal{S},\mathcal{U}_1) \cup \mathrm{S}^R(\mathcal{S},\mathcal{U}_2)$.
(·) *If* $\mathrm{w}^R(\mathcal{S}_j,\mathcal{U}_j) \neq \infty$ *for each* $j \in [2]$, *then there exists some* $r \in R$ *such that* $\mathrm{S}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) \subseteq \{r\}$ $(\subseteq \mathrm{S}^R(\mathcal{S}_1,\mathcal{U}_1) \cup \mathrm{S}^R(\mathcal{S}_2,\mathcal{U}_2) \cup \{r\})$.

Proof sketch. (WK): Trivial. (At): By that R is good for PNF-PCoR (see also Lemma 6.4). (∪) and (∩): They have already been proved in Lemma 5.7. (·): In the same way as the proof of Lemma 5.10. If we assume that such r does not exist, then $\mathrm{w}^R(\mathcal{S}_1 \cdot \mathcal{S}_2, \mathcal{U}_1 \dagger \mathcal{U}_2) \geq 2$ should. Thus by Lemma 5.9 (3), $\mathrm{w}^R(\mathcal{S}_j,\mathcal{U}_j) = \infty$ for some $j \in [2]$, but it reaches a contradiction. Hence, this completes the proof. □

Using Lemma 6.14, we can show the following lower bound for $\| \bullet \|_{(\mathrm{DAG})}$.

Lemma 6.15. *Let t be a PNF-NNF-*Σ_1^{CoR} *term and let $\mathfrak{T}$ be a derivation tree of $\langle \mathcal{S},\mathcal{U}\rangle \vdash t$. Assume that R is good for PNF-*PCoR.

(1) *For every $r \in \mathrm{S}^R(\mathcal{S},\mathcal{U})$, there exists some occurrence*
$$\frac{\langle \mathcal{S}_1^{(r)},\mathcal{U}_1^{(r)}\rangle \vdash t_1^{(r)} \quad \langle \mathcal{S}_2^{(r)},\mathcal{U}_2^{(r)}\rangle \vdash t_2^{(r)}}{\langle \mathcal{S}_1^{(r)} \cdot \mathcal{S}_2^{(r)}, \mathcal{U}_1^{(r)} \dagger \mathcal{U}_2^{(r)}\rangle \vdash t_1^{(r)} \cdot t_2^{(r)}}$$
in $\mathfrak{T}$ such that $r \in \mathrm{S}^R(\mathcal{S}_1^{(r)} \cdot \mathcal{S}_2^{(r)}, \mathcal{U}_1^{(r)} \dagger \mathcal{U}_2^{(r)}) \setminus (\mathrm{S}^R(\mathcal{S}_1^{(r)},\mathcal{U}_1^{(r)}) \cup \mathrm{S}^R(\mathcal{S}_2^{(r)},\mathcal{U}_2^{(r)}))$.
(2) *In (1), for every distinct $r_1, r_2 \in \mathrm{S}^R(\mathcal{S},\mathcal{U})$, $t_1^{(r_1)} \cdot t_2^{(r_1)} \neq t_1^{(r_2)} \cdot t_2^{(r_2)}$.*
(3) *t has at least $\mathrm{w}^R(\mathcal{S},\mathcal{U})$ subterms, hence $\|t\|_{(\mathrm{DAG})} \geq \mathrm{w}^R(\mathcal{S},\mathcal{U})$.*

Proof. (1): If we assume that such occurrence does not exist, then we can show $r \notin \mathrm{S}^R(\mathcal{S},\mathcal{U})$ by induction on the structure of $\mathfrak{T}$ using Lemma 6.14. Thus, it reaches a contradiction. (2): If we assume $t_1^{(r_1)} \cdot t_2^{(r_1)} = t_1^{(r_2)} \cdot t_2^{(r_2)}$, then let $t_i = t_i^{(r_1)} = t_i^{(r_2)}$ for each $i \in [2]$. By $r_i \in \mathrm{S}^R(\mathcal{S}_1^{(r_i)} \cdot \mathcal{S}_2^{(r_i)}, \mathcal{U}_1^{(r_i)} \dagger \mathcal{U}_2^{(r_i)})$ and Lemma 5.9(2), $\mathrm{w}^R(\bigcup_{i\in[2]} \mathcal{S}_j^{(r_i)}, \bigcup_{i\in[2]} \mathcal{U}_j^{(r_i)}) = \infty$ for some $j \in [2]$. Also, by $\langle \mathcal{S}_j^{(r_i)}, \mathcal{U}_j^{(r_i)}\rangle \vdash t_j$ for each $i \in [2]$ and the completeness (Theorem 4.3), $\langle \bigcup_{i\in[2]} \mathcal{S}_j^{(r_i)}, \bigcup_{i\in[2]} \mathcal{U}_j^{(r_i)}\rangle \vdash t_j$. Therefore by Proposition 4.6, we have $\|t_j\| \geq \infty$. Thus, it reaches a contradiction since such t_j does not exist. Hence, $t_1^{(r_1)} \cdot t_2^{(r_1)} \neq t_1^{(r_2)} \cdot t_2^{(r_2)}$. (3): By (2). □

Lemma 6.16 *(cf. Proposition 4.6). Assume that R is good for PNF-*PCoR. *for every PNF-NNF-*Σ_1^{CoR} *term t, $\langle \mathcal{S},\mathcal{U}\rangle \vdash t \Longrightarrow \|t\|_{(\mathrm{DAG})} \geq \mathrm{w}^R(\mathcal{S},\mathcal{U})$. Thus, $\mathrm{sz}_{\text{PNF-NNF-}\Sigma_1^{\mathrm{CoR}}{}_{(\mathrm{DAG})}}(\mathcal{S},\mathcal{U}) \geq \mathrm{w}^R(\mathcal{S},\mathcal{U})$.*

Proof. By Lemma 6.15(3). □

Proof of Lemma 6.13. Using Lemma 6.16 (instead of Proposition 4.6), and using the same sequence $\langle \mathcal{S}^{(n)},\mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ in Definition 5.13 and the same weight function $\mathrm{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}$, this completes the proof. □

7. On the transitive closure extension

In this section, we show that the results of the expressive power equivalence (Section 3) and the succinctness (Section 5) can be extended to the positive calculus of relations with transitive closure [9] (denoted by PCoR(TC), for short). We will show that the calculus has the same expressive power as three-variable existential positive (first-order) logic with (variable-confined) monadic transitive closure (denoted by EP^3(v-MTC)) (see, e.g., [17, Sect. 9] for transitive closure logic). For a binary relation R and $k \in \mathbb{N}_+$, we use $R^{k\cdot}$ to denote $R \cdot R^{(k-1)\cdot}$ if $k \geq 2$; and R if $k = 1$. Similarly, we use $R^{k\dagger}$ to denote $R \dagger R^{(k-1)\dagger}$ if $k \geq 2$; and R if $k = 1$. Also, we use R^+ to denote $\bigcup_{k\in\mathbb{N}_+} R^{k\cdot}$ and use $R^{\oplus}$ to denote $\bigcap_{k\in\mathbb{N}_+} R^{k\dagger}$.

The *syntax* of PCoR(TC) is given by:

$$t, s ::= a \mid \mathbf{1} \mid \top \mid \mathbf{0} \mid t \cup s \mid t \cap s \mid t^{\pi} \mid t \cdot s \mid t^+.$$

The *size* $\|t\|$ and the *semantics* $[\![t]\!]_M$ are defined in the same way as for CoR, respectively, where $\|t^+\| := 1 + \|t\|$ and $[\![t^+]\!]_M := [\![t]\!]^+_M$.

Also, the *syntax* of EP(v-MTC) is given by the following grammar, where $x, y, u, w \in \mathscr{V}$; u and w are distinct; and each $[\varphi]^+_{u,w}$ is *variable-confined* (i.e., $\mathbf{FV}(\varphi) \subseteq \{u, w\}$[3]):

$$\varphi, \psi ::= a(x, y) \mid x = y \mid \mathsf{tt} \mid \mathsf{ff} \mid \varphi \vee \psi \mid \varphi \wedge \psi \mid \exists x.\varphi \mid [\varphi]^+_{u,w}(x, y).$$

The *size* $\|\varphi\|$ and the *semantics* $\langle M, I\rangle \models \varphi$ are defined in the same way as for EP, where $\|[\varphi]^+_{u,w}(x, y)\| := 1 + \|\varphi\|$ and $\langle M, I\rangle \models [\varphi]^+_{u,w}(x, y) :\Leftrightarrow \langle I(x), I(y)\rangle \in \bigcup_{k\in\mathbb{N}_+} [\![[\varphi]_{u,w}]\!]^k_M$.

Then, the following holds by using the same strategy as that in Section 3.

Theorem 7.1. PCoR(TC) $=_{\mathbf{M}_2}$ EP^3(v-MTC). *More precisely,*

(1) PCoR(TC) $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ EP^3(v-MTC).
(2) EP^3(v-MTC) $\leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}$ PCoR(TC).

Proof sketch. (1): By extending the standard translation $\mathsf{ST}_{x,y}$ for $\bullet^+$ by: $\mathsf{ST}_{x,y}(t^+) := [\mathsf{ST}_{x,y}(t)]^+_{x,y}(x, y)$. (2): Let us recall the translations (‡2)-(‡4). (‡2): The translation in (‡2) does work by extending the translation $\mathsf{T}_\bullet$ as follows: $\mathsf{T}_-(\varphi) := [\mathsf{T}_-(\psi)]^+_{u,w}(x, y)$ and $\mathsf{T}_\forall(\varphi) := \mathsf{T}_\exists(\varphi) := \langle\langle[\mathsf{T}_-(\psi)]^+_{u,w}(x, y)\rangle\rangle$ if $\varphi \equiv [\psi]^+_{u,w}(x, y)$. Hence, EP^3(v-MTC) $\leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}$ good-EP^3(v-MTC). (‡3): For distinct x and y, we say that an EP^3(v-MTC) formula is ($\{x, y\}$-)nice if it is in the set defined by the following grammar, where $\tilde{x}, \tilde{y} \in \{x, y\}$ and z is the variable distinct from x and y:

$$\varphi^{\{x,y\}}, \psi^{\{x,y\}} ::= a(\tilde{x}, \tilde{y}) \mid \tilde{x} = \tilde{y} \mid \mathsf{tt} \mid \mathsf{ff} \mid [\varphi^{\{u,w\}}]^+_{u,w}(\tilde{x}, \tilde{y}) \mid \varphi^{\{x,y\}} \vee \psi^{\{x,y\}} \mid \varphi^{\{x,y\}} \wedge \psi^{\{x,y\}} \mid \exists z.\varphi^{\{x,z\}} \wedge \psi^{\{z,y\}}$$

Then good-EP^3(v-MTC) $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ nice-EP^3(v-MTC) by using the same translation in (‡3). (‡4): The translation T in (‡4) does work by extending it as follows: $\mathsf{T}([[\varphi^{\{u,w\}}]^+_{u,w}(x_i, x_j)]_{x_1,x_2}) := (\mathsf{T}([\varphi^{\{u,w\}}]_{u,w})^+)^{\{1\mapsto i, 2\mapsto j\}}$. From this, nice-EP^3(v-MTC) $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ PCoR(TC). Therefore, EP^3(v-MTC) $\leq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}$ PCoR(TC) by combining (‡2)-(‡4). □

Also, for the succinctness, the following holds.

Theorem 7.2. EP^3 $\not\leq^{2^{o(n)}}_{\mathbf{M}_2}$ PCoR(TC).

The proof is as with the proof of Theorem 5.1 by extending each notion for $\bullet^+$. First, we extend the formula-size game for $\bullet^+$. For a structure class $\mathcal{C} \subseteq \mathbf{M}_2$ and $k \in \mathbb{N}_+$, we use $\mathcal{C}^{k\cdot}$ to denote $\mathcal{C} \cdot \mathcal{C}^{(k-1)\cdot}$ if $k \geq 2$ and $\mathcal{C}$ if $k = 1$. Similarly, we use $\mathcal{C}^{k\dagger}$ to denote $\mathcal{C} \dagger \mathcal{C}^{(k-1)\dagger}$ if $k \geq 2$ and $\mathcal{C}$ if $k = 1$. Also, we use $\mathcal{C}^+$ to denote $\bigcup_{k\in\mathbb{N}_+} \mathcal{C}^{k\cdot}$ and use $\mathcal{C}^\oplus$ to denote $\bigcap_{k\in\mathbb{N}_+} \mathcal{C}^{k\dagger}$. Then we extend the formula-size game in Fig. 4 by adding the following rule:

$$\frac{\langle \mathcal{S}_1, \mathcal{U}_1\rangle \vdash t}{\langle \mathcal{S}_1^+, \mathcal{U}_1^\oplus\rangle \vdash t^+}\ (\bullet^+)$$

Then the following holds.

Lemma 7.3.

($\bullet^+$) $\langle \mathcal{S}, \mathcal{U}\rangle \models t^+$ *iff for some* $\mathcal{S}_1, \mathcal{U}_1$ *such that* $\mathcal{S}_1 \subseteq \mathcal{S}_1^+$ *and* $\mathcal{U}_1 \subseteq \mathcal{U}_1^\oplus$, $\langle \mathcal{S}_1, \mathcal{U}_1\rangle \models t$.

Proof sketch. As with Lemma 4.2, it is straightforward by the definition of $\models$. □

Theorem 7.4 *(sound- and completeness). For all* PCoR(TC) *term* t, $\langle \mathcal{S}, \mathcal{U}\rangle \vdash t$ *iff* $\langle \mathcal{S}, \mathcal{U}\rangle \models t$.

Proof sketch. As with Theorem 4.3 using Lemma 7.3. □

[3] Here, z and u in $\mathrm{TC}_{u,w}(\psi)$ are viewed as bound variables (i.e., $\mathbf{FV}([\mathrm{TC}_{u,w}(\psi)](x, y))$ is defined by $\mathbf{FV}([\mathrm{TC}_{u,w}(\psi)](x, y)) := (\mathbf{FV}(\psi) \setminus \{u, w\}) \cup \{x, y\}$). See also [17, Sect. 9].

Secondly, we define the set of *projection normal form* PCoR(TC) as with PNF-PCoR. Then the following holds.

Proposition 7.5. PCoR(TC) $\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}$ *PNF*-PCoR(TC).

Proof sketch. As with Proposition 5.2 using the rewriting rule $(t^+)^\smile \rightsquigarrow (t^\smile)^+$. □

From this, it suffices to show the following (as with Lemma 5.3).

Lemma 7.6. $\mathrm{EP}^3 \not\leq^{2^{o(n)}}_{\mathbf{M}_2}$ *PNF*-PCoR(TC).

We extend the weight function for $\bullet^+$ as follows.

Definition 7.7. We say that a function $\mathrm{w}\colon 2^{\mathbf{M}_2} \times 2^{\mathbf{M}_2} \to \mathbb{N} \cup \{\infty\}$ is *a weight function for PNF*-PCoR(TC) if all the conditions of weight function for PNF-PCoR (Definition 4.5) and the following holds:

($\bullet^+$) $\quad \mathrm{w}(\mathcal{S}_1^+, \mathcal{U}_1^\oplus) \leq \mathrm{w}(\mathcal{S}_1, \mathcal{U}_1) + 1.$

We extend the notion "good for PNF-PCoR" for PCoR(TC) as follows.

Definition 7.8. We say that $R \subseteq \mathbf{M}_2^2$ is *good for PNF*-PCoR(TC) if all the conditions of "good for PNF-PCoR" (Definition 5.8) and the following holds:

($\bullet^+$) For every $\langle\langle M, v, v'\rangle, \langle N, w, w'\rangle\rangle \in R$, there is w'' such that $\langle M, v, v'\rangle \subseteq \langle N, w, w''\rangle$ and $\langle M, v, v'\rangle \subseteq \langle N, w'', w'\rangle$.

Lemma 7.9. *If R is good for PNF*-PCoR(TC)*, then the conditions (At) and (·) in Lemma 5.10 and the following hold:*

($\bullet^+$) $\quad \mathrm{w}^R(\mathcal{S}_1^+, \mathcal{U}_1^\oplus) \geq 1 \Longrightarrow \mathrm{w}^R(\mathcal{S}_1, \mathcal{U}_1) = \infty.$

Hence, w^R *is a weight function for PNF*-PCoR(TC).

Proof. Case $\mathrm{w}^R(\mathcal{S}_1^+, \mathcal{U}_1^\oplus) = \infty$: Let $\langle\langle M, v_0, v'\rangle, \langle N, w_0, w'\rangle\rangle \in \mathcal{S}_1^+ \times \mathcal{U}_1^\oplus$ be a substructure pair and let $h\colon \langle M, v_0, v'\rangle \subseteq \langle N, w_0, w'\rangle$ be an embedding. Then, there exist $v_1, \ldots, v_n$ such that $\langle M, v_{i-1}, v_i\rangle \in \mathcal{S}_1$ for every $i \in [n]$ and $v_n = v'$. Then by $\langle N, w_0, w'\rangle \in \mathcal{U}_1^\oplus$, $\langle N, h(v_{i-1}), h(v_i)\rangle \in \mathcal{U}_1$ holds for some $i \in [n]$. Therefore, there exists some substructure pair in $\mathcal{S}_1 \times \mathcal{U}_1$, and hence $\mathrm{w}^R(\mathcal{S}_1, \mathcal{U}_1) = \infty$.

Otherwise: Let $\langle\langle M, v_0, v'\rangle, \langle N, w_0, w'\rangle\rangle \in ((\mathcal{S}_1^+)^\cong \times (\mathcal{U}_1^\oplus)^\cong) \cap R$. Then, there exist $v_1, \ldots, v_n$ such that $\langle M, v_{i-1}, v_i\rangle \in \mathcal{S}_1^\cong$ for every $i \in [n]$ and $v_n = v'$. By Definition 7.8($\bullet^+$), there exist w'' and embeddings $h_1\colon \langle M, v_0, v_n\rangle \subseteq \langle N, w_0, w''\rangle$ and $h_2\colon \langle M, v_0, v_n\rangle \subseteq \langle N, w'', w'\rangle$. Then by $\langle N, w_0, w'\rangle \in (\mathcal{U}_1^\oplus)^\cong$, one of the following holds: $\langle N, h_1(v_0), h_1(v_1)\rangle \in \mathcal{U}_1^\cong$, …, $\langle N, h_1(v_{n-1}), h_1(v_n)\rangle \in \mathcal{U}_1^\cong$, $\langle N, h_2(v_0), h_2(v_1)\rangle \in \mathcal{U}_1^\cong$, …, $\langle N, h_2(v_{n-1}), h_2(v_n)\rangle \in \mathcal{U}_1^\cong$. Hence, there exist a substructure pair in $\mathcal{S}_1^\cong \times \mathcal{U}_1^\cong$ (so also in $\mathcal{S}_1 \times \mathcal{U}_1$).

Combining with Lemma 5.11, w^R is a weight function for PNF-PCoR(TC). □

Using this lemma, it suffices to show the following to prove Lemma 7.6 as with Lemma 5.12.

Lemma 7.10. *There exists a sequence $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle_{n\in\mathbb{N}_+}$ such that*

(1) *$\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}$ is good for PNF*-PCoR(TC)*, and*
(2) *$\langle \mathrm{sz}_{\mathrm{EP}^3}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \in \mathcal{O}(n)$,*
(3) *$\langle \#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \not\leq_{\mathrm{b}} 2^{o(n)}$.*

By using Lemma 5.12, Lemma 5.4 is shown as follows.

Proof of Lemma 7.10 $\Longrightarrow$ Lemma 7.6. Let us assume the following:
$\langle \mathrm{sz}_{\text{PNF-PCoR(TC)}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \leq_{\mathrm{b}} 2^{o(n)}$. Then it contradicts (3) as follows:

$$\begin{aligned}\langle \#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} &= \langle \mathrm{w}^{\mathcal{S}^{(n)}\times\mathcal{U}^{(n)}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \\ &\leq_{\mathrm{b}} \langle \mathrm{sz}_{\text{PNF-PCoR(TC)}}(\mathcal{S}^{(n)}, \mathcal{U}^{(n)})\rangle_{n\in\mathbb{N}_+} \leq_{\mathrm{b}} 2^{o(n)}.\end{aligned}$$

Therefore by Proposition 4.7 with $\mathcal{F} = \mathcal{O}(n)$ and $\mathcal{G} = 2^{o(n)}$ (note that $2^{o(n)} \circ \mathcal{O}(n) = 2^{o(n)\circ\mathcal{O}(n)} \leq_{\mathrm{b}} 2^{o(n)}$). □

Finally, we construct a sequence of sets of structures for Lemma 7.10. Let us recall $\tilde{\mathbb{S}}_I^{(n)}$ and $\tilde{\mathbb{U}}^{(n)}$ in Definition 5.13. Then, we define $(\tilde{\mathbb{U}}')^{(n)}$ as follows.

$((\tilde{\mathbb{S}}_I^2)^{(n)})$ Let $n \in \mathbb{N}$ and $I \subseteq [n]$. Then, let $(\tilde{\mathbb{S}}_I^2)^{(n)}$ be the 2-pointed structure $\langle (\mathbb{S}_I^2)^{(n)}, \mathsf{L}, \mathsf{R}\rangle$ defined by: $(\mathbb{S}_I^2)^{(n)} := (\mathbb{S}_I^{(n)} \uplus \mathbb{S}_I^{(n)})/\sim$ and $\langle \mathsf{L}, \mathsf{R}\rangle := \langle [\langle 1, \mathsf{L}\rangle]_\sim, [\langle 2, \mathsf{R}\rangle]_\sim\rangle$. Here, $\sim$ is the minimum equivalence relation satisfying $\langle 1, \mathsf{R}\rangle \sim \langle 2, \mathsf{L}\rangle$.

$((\tilde{\mathbb{U}}')^{(n)})$ Let $n \in \mathbb{N}$ and let K be the set of all the pairwise distinct I_k such that $\emptyset \subsetneq I_k \subsetneq [n]$. Then, let $(\tilde{\mathbb{U}}')^{(n)}$ be the 2-pointed structure $\langle (\mathbb{U}')^{(n)}, \mathsf{L}, \mathsf{R}\rangle$ defined as follows: $(\mathbb{U}')^{(n)} := (\mathbb{U}^{(n)} \uplus (\biguplus_{k\in[\#K]} (\mathbb{S}_{I_k}^2)^{(n)}))/\sim$ and $\langle \mathsf{L}, \mathsf{R}\rangle := \langle [\langle 1, \mathsf{L}\rangle]_\sim, [\langle 1, \mathsf{R}\rangle]_\sim\rangle$. Here, $\sim$ is the minimum equivalence relation satisfying $\langle 1, \mathsf{L}\rangle \sim \langle 2, \langle i, \mathsf{L}\rangle\rangle$ and $\langle 1, \mathsf{R}\rangle \sim \langle 2, \langle i, \mathsf{R}\rangle\rangle$.

Then, we let $\mathcal{S}^{(n)} := \{\tilde{\mathbb{S}}_I^{(n)} \mid \emptyset \subsetneq I \subsetneq [n]\}$ and $\mathcal{U}^{(n)} := \{(\tilde{\mathbb{U}}')^{(n)}\}$.

Proof of Lemma 7.10. (1): (At), $(\bullet^{\{1\mapsto 1,2\mapsto 1\}})$, $(\bullet^{\{1\mapsto 2,2\mapsto 2\}})$, and $(\cdot)$ are shown in the same way as Proposition 5.14 since $(\tilde{\mathbb{U}}')^{(n)}$ is a substructure of $\tilde{\mathbb{U}}^{(n)}$. $(\bullet^+)$ is shown by using the part of $(\tilde{\mathbb{S}}_I^2)^{(n)}$ in $(\tilde{\mathbb{U}}')^{(n)}$. (2): By $\langle \mathcal{S}^{(n)}, \mathcal{U}^{(n)}\rangle \models \mathfrak{t}_n$ and $\langle \|\mathfrak{t}_n\|\rangle_{n\in\mathbb{N}_+} \in \mathcal{O}(n)$. Here, $\mathfrak{t}_n$ is the EP^3 term presented in Section 5. (3): By $\#(\mathcal{S}^{(n)} \times \mathcal{U}^{(n)}) = 2^n - 2$. □

8. Conclusion

We have shown that (1) the positive calculus of relations has the same expressive power as three-variable existential positive logic, and (2) the positive calculus of relations is exponentially less succinct than three-variable existential positive logic. To the best of our knowledge, it is open whether the calculus of relations is exponentially less succinct than three-variable first-order logic. Table 1 summarizes the expressive power and the succinctness results between fragments of CoR and FO^3. (In Table 1, $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$? denotes that for the term class $\mathfrak{T}$ in the row label the term class $\mathfrak{S}$ in the column label, it is open whether or not $\mathfrak{T} \not\geq^{2^{o(n)}}_{\mathbf{M}_2} \mathfrak{S}$.) Similarly, Table 2 summarizes the ones between fragments of CoR^2 and FO^2. Here, in the tables, the set of binary relation symbols is countably infinite and the (tree-)size $\|\bullet\|$ is used (the succinctness results can be extended for the DAG-size $\|\bullet\|_{(\mathrm{DAG})}$; see Section 6.3). It would also be interesting to construct a calculus like the

Table 1
Summary between fragments of CoR and FO^3.

	PP^3	EP^3	Σ_1^3 (Definition 3.12)	Σ_n^3 $(n \geq 2)$ (Definition 3.12)	FO^3 $=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ NNF-FO^3 (Sect. 3)
PPCoR	$=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ (Corollary 3.2)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$ (Proposition 3.18)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$
PCoR $=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ PNF-PCoR (Sect. 5)	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$ (Proposition 3.18)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ (Theorem 2.8 [2]) (Theorem 3.1) $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$ (Proposition 3.17)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$
Σ_1^{CoR} (Definition 3.13) $=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ PNF-NNF-Σ_1^{CoR} (Sect. 6.1)	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ (Proposition 3.17) $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$ (Theorem 6.1)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ (Corollary 3.14 [2]) (Corollary 3.14) $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$ (Proposition 3.15 [29])	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$
Σ_n^{CoR} $(n \geq 2)$ (Definition 3.13)	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$?	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ (Proposition 3.15 [29]) $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$?	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ (Corollary 3.14 [2]) (Corollary 3.14) $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$?	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$ (Proposition 3.15 [29])
CoR	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$?	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$?	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$?	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\geq^{\mathcal{O}(2^n)}_{\mathbf{M}_2}}$ (Theorem 2.8 [2]) (Theorem 3.1) $\not\geq^{2^{o(n)}}_{\mathbf{M}_2}$?

Table 2
Summary between fragments of CoR^2 and FO^2.

	PP^2	EP^2	Σ_n^2 $(n \geq 1)$ (Sect. 3.2)	FO^2
PPCoR^2	$=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ (Corollary 3.20)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$ (Proposition 3.24)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$
PCoR^2	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$ (Proposition 3.24)	$=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ (Corollary 3.20)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$ (Proposition 3.23)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$
$\Sigma_n^{\mathrm{CoR}^2}$ $(n \geq 1)$ (Sect. 3.2)	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$ (Proposition 3.23)	$=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ (Corollary 3.21)	$\dfrac{\leq^{\mathcal{O}(n)}_{\mathbf{M}_2}}{\not\geq_{\mathbf{M}_2}}$ (Corollary 3.21)
CoR^2	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$	$\dfrac{\not\leq_{\mathbf{M}_2}}{\geq^{\mathcal{O}(n)}_{\mathbf{M}_2}}$ (Corollary 3.21)	$=^{\mathcal{O}(n)}_{\mathbf{M}_2}$ (Corollary 3.20)

(positive) calculus of relations (or cylindric algebra [30]) such that it has the same expressive power as k-variable (existential positive) first-order logic, and there is a succinctness gap between them.

Declaration of competing interest

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